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**Intergroup Leadership in Multiteam Systems: The Effect of Identification,
Systems Thinking, and Functional Diversity on**

Intergroup Leadership and Multiteam System Effectiveness

A Dissertation

Submitted to the Faculty

of

Drexel University

by

Theresa Cecilia Carter

in partial fulfillment of the

requirements for the degree

of

Doctor of Business Administration

June 2021



The views, opinions, and/or findings contained in this dissertation are those of the author and should not be construed as an official Department of the Air Force position, policy, or decision.

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DEDICATIONS

To all of the men and women with whom I served for 31 years in the United States Air Force – it was my honor and privilege to serve with you. Each of you in some way served as inspiration for this research as I reflected on our collective successes and the times when my own leadership fell short of the mark. I hope the findings from this study and future research in this area help build leaders who are equipped to handle the challenges associated with leading a team of teams in complex and dynamic environments.

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ABSTRACT

Intergroup leadership in multiteam systems: The effect of identification, systems thinking, and functional diversity on intergroup leadership and multiteam system effectiveness

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Even though empirical studies examining leadership and effectiveness in multiteam systems (MTS) are on the rise, there is still much to learn about leading a complex “team of teams,” particularly in a field setting as opposed to a laboratory simulation. This study draws upon the multiteam systems and intergroup leadership theory literature streams to examine the role of systems thinking and leader identification as antecedents to intergroup leadership, arguing that higher levels of systems thinking and stronger leader identification with the MTS will enhance the efficacy of leader rhetoric targeted toward the development of an intergroup relational identity with boundary spanning acting as a positive moderating influence on this relationship. I further argue team functional diversity moderates the relationship between intergroup leadership and MTS effectiveness such that higher levels of intergroup leadership are needed to positively influence MTSs with higher levels of heterogeneity. Finally, team identification is evaluated for its predicted moderating effect on the relationship between intergroup leadership and MTS effectiveness such that identification with the MTS will have a positive effect on this relationship while identification with the component team will detract from it. Hypotheses were tested but not supported using a sample of 256 commanders at the squadron, group, and wing level organized in 180 MTSs within the

Department of the Air Force. Supplemental analysis identified a significant positive moderating effect of systems thinking and a negative moderating effect of leader component team identification on the relationship between leader rhetoric and intergroup relational identity. Low levels of boundary spanning coupled with higher levels of intergroup relational identity resulted in lower MTS effectiveness while the opposite was true for high levels of boundary spanning and intergroup relational identity. The study concludes with a discussion of theoretical and practical implications as well as recommendations for future research.

KEY WORDS: Multiteam systems, Intergroup leadership, Systems thinking, Identification, Military

CHAPTER 1: INTRODUCTION

Introduction to the Problem

Two decades have passed since Mathieu, Marks, and Zaccaro (2001) published their seminal work on multiteam systems (MTS), a new organizational framework intended to capture the complex environment in which multiple interdependent teams work in concert to achieve a shared superordinate goal. The context in which an MTS operates can range from emergency responders treating patients at an accident scene to complex interagency task forces managing the aftermath of a natural disaster, a military humanitarian relief operation, or complex collaborations involving research and development teams from multiple firms (Zaccaro, Dubrow, Torres, & Campbell, 2020).

The core factors distinguishing an MTS from other types of collectives and groups are team interdependence (input, process, and outcome) and goal hierarchies that include proximal team and distal system goals (Zaccaro et al., 2020). The superordinate goal and interdependence between component teams characterize the extent of the MTS boundaries rather than an organizational border or wiring diagram (Mathieu, 2012). An MTS by design goes beyond what any one team can achieve, leveraging the unique skills and expertise resident in component teams and synchronizing their actions toward superordinate goal accomplishment (Shuffler, Rico, & Salas, 2014).

Although the volume of MTS research is growing and we have a better understanding today about MTS dynamics than two decades ago, there is much left to learn and discover (Mathieu, Luciano, & DeChurch, 2018; Waring, 2019; Zaccaro et al., 2020). Zaccaro and colleagues (2020) reviewed 115 empirical papers published since

2001 and evaluated findings in three areas: within- and between-team coordination, leadership, and cognitive and affective emergent states. They categorized studies by the complexity of boundary status (internal or external), component team distance (location and discipline diversity), and goal type (intellectual or physical), finding external MTSs with high component team distance displayed high differentiation and likely needed higher levels of interpersonal processes such as coordination for MTS success. Other recent conceptual papers and reviews of previous work include Waring's (2019) review of studies using exercises or simulations focused on methodological and measurement fit, Mathieu and colleagues' (2018) examination of the first fifteen years of MTS research, a review of studies addressing teamwork in MTSs (Shuffler & Carter, 2018), a review of teams research published in the *Journal of Applied Psychology* over the past century (Mathieu, Hollenbeck, van Knippenberg, & Ilgen, 2017), an examination distilling the state of the science regarding MTSs focused on what we know and where we need to go (Shuffler, Jimenez-Rodriguez, & Kramer, 2015), and a historiometric analysis of critical incident after action reports focusing on leadership issues (DeChurch, Burke, Shuffler, Lyons, Doty, & Salas, 2011).

Much of the early MTS research used computer simulations to evaluate MTS attributes such as leadership or coordination. Waring (2019) provided a comprehensive summary of studies using these simulations or exercises. Small scale combat flight simulations (Air Combat Effectiveness Simulation) with undergraduate students were used to assess the impact of environmental complexity and team training (Cobb, 1999) and component team coordination (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005) on MTS performance. Bienefeld and Grote (2014) employed a flight crew simulation

exercise to evaluate shared leadership. Multiple studies utilized the leadership development simulation (LDS) with teams of Air Force captains attending Squadron Officer School (SOS) as the study population. Davison, Hollenbeck, Barnes, Sleesman, & Ilgen (2012) evaluated horizontal and vertical coordination within and between component teams and its impact on MTS performance while DeVries, Hollenbeck, Davison, Walter, and van der Vegt (2016) examined the relationship between vertical and horizontal coordination, intrapersonal functional diversity, and MTS performance. Reducing representational gaps and understanding their impact on coordination and MTS performance (Firth, Hollenbeck, Miles, Ilgen, & Barnes, 2015) as well as understanding how decentralized planning impacts MTS performance (Lanaj, Hollenbeck, Ilgen, Barnes, & Harmon, 2013) were also studied using the LDS platform at SOS. DeChurch and colleagues (2011) used historiometric analysis of after-action reports to examine leader behaviors and functions when the MTS is situated in an extreme environment (e.g., disaster response or combat zone).

While computer simulations allow researchers to precisely measure performance outcomes and manipulate other variables, it is important to avoid having the simulation be the focus rather than the study of the MTS (Mathieu et al., 2018; Resick, Burke, & Doty, 2012). Measures of MTS effectiveness in laboratory settings often rely on metrics built into the simulation program, challenging external validity of the findings in other settings or contexts (Resick et al., 2012). Shuffler and colleagues (2014) argue there may be issues with MTSs “in the wild” that laboratory settings cannot capture or emulate, and multiple authors call for an increase in research that focuses on MTS dynamics in the field (Davison et al., 2012; Mathieu, 2012; Shuffler et al., 2014; Waring, 2019).

To address these gaps in knowledge, improve understanding of leadership in an MTS, and assess its impact on MTS effectiveness, this study relies upon the emerging MTS literature stream to examine leadership in a complex MTS, specifically examining leadership through the lens of intergroup leadership theory. This study makes a valuable contribution to both the researcher and practitioner communities as it is conducted as a field study across multiple contextual environments and examines MTS leadership and effectiveness in a nested hierarchical structure. Uncovering drivers and influences on intergroup leadership and MTS effectiveness will also inform researchers where to concentrate their focus in future studies and practitioners will better understand how to improve or create new leadership development programs aimed at preparing individuals to successfully lead these complex entities.

Purpose and Significance of the Problem

As Aristotle astutely observed, *“The more you know, the more you know you don’t know.”* While the growing interest in multiteam systems and intergroup leadership is helping drive an increase in conceptual papers and empirical studies, the number of scholarly journal articles and dissertations remains extremely limited. Research has only begun to scratch the surface of the myriad facets involved in MTS leadership and performance. While the findings from most empirical studies to date have been generally in agreement surrounding the strength and direction of influences on leadership and performance, contradictory and surprising findings show how much researchers have yet to discover. Most of the intergroup leadership theory propositions advanced by Hogg, van Knippenberg, and Rast (2012) lack empirical testing and it is essential to conduct research in this area to further our understanding of the challenge leaders face in today’s

complex organizations – how do you effectively lead and exert influence between teams and across system boundaries to achieve effective system performance?

From a practitioner perspective, understanding the drivers of effective intergroup leadership and MTS performance will allow organizations to design leadership development programs focused on building the knowledge, skills, and abilities required to lead across groups and within these complex systems. Intergroup leadership theory (Hogg et al., 2012), while still relatively new, is grounded in social identity and intergroup relations theory and empirical studies using lab and field investigations are underway to test its propositions (Rast, Hogg, & van Knippenberg, 2018; Porck & van Knippenberg, 2020; Kershaw, Rast, Hogg, & van Knippenberg, 2021a; Kershaw, Rast, Hogg, & van Knippenberg, 2021b; van der Stoep, Sleafos, van Knippenberg, & Bunt, 2020). Intergroup leadership theory's components and concepts can form the basis for a common frame of reference in researcher and practitioner communities and serve as a foundation for individual and team training programs centered on intergroup leadership and leadership in an MTS. Identifying and empirically testing leader knowledge, skills, and abilities (KSAs) as antecedents to intergroup leadership will inform existing or aid in the creation of new leader training and development programs that promote an understanding of leading within complex systems and in turn equip leaders to exercise intergroup leadership to improve system effectiveness. As Hogg and colleagues (2012) argue, empirical studies are needed to inform organizations regarding their leader development programs. "The skills needed to convey the appropriate intergroup relational message by using compelling rhetoric and building boundary-spanning relationships may not be easy to master" (Hogg et al., 2012: 250).

Thus, the purpose of this study is to examine intergroup leadership in the context of a complex MTS in a field setting, evaluate the influence of specific leader knowledge, skills, and abilities that may promote intergroup leadership and MTS effectiveness, and understand how functional diversity and social identification at the individual, component team, and MTS levels influence the relationship between intergroup leadership and MTS effectiveness. Leveraging the MTS and intergroup leadership theory literature streams, this study will examine intergroup leadership in an MTS through a quantitative study examining squadrons, groups, and wings in the Department of the Air Force.

Research Questions

My aim with this study is to address three research questions which to date have either limited or no empirical evaluation in the literature. First, *how does boundary spanning facilitate or hinder intergroup leadership and in turn influence multiteam system effectiveness?* Empirical support for boundary spanning leadership's positive effect on team and system performance in general (Ancona & Caldwell, 1992; Davison et al., 2012; Ernst & Yip, 2009; Faraj & Yan, 2009; Korschun, 2015; Marrone, Tesluk, & Carson, 2007; Marrone, 2010; Richter, West, Van Dick, & Dawson, 2006) highlights the importance of considering this behavior when examining leadership in an MTS.

Boundary spanners are often part of a component team or leadership team and act as team representatives as well as collectors and distributors of knowledge gained through their boundary spanning activities (Ancona, 1990; Davison et al., 2012; Firth et al., 2015; Lanaj et al., 2103). DeChurch & Marks (2006) observed that one of the most essential leadership functions in an MTS is acting as a bridge between teams and providing an overall vision and understanding of MTS resources, constraints, and performance

measures. Boundary spanning leadership promotes collaboration and cooperation between teams and in turn a sense of belonging among team members such that component teams and the MTS as whole operate more effectively (DeCostanza, DiRosa, Jiménez-Rodriguez, & Cianciolo, 2014; Shuffler & Carter, 2018).

In the specific context of intergroup leadership, Hogg and colleagues (2012) offered two intergroup leadership theory propositions with respect to boundary spanning. First, boundary spanning leadership acts a moderator, enhancing the impact of leader rhetoric aimed at building an intergroup relational identity. Extending this individual leadership activity to the group level, they next argue “Intergroup leadership targeted at engendering intergroup relational identity is more effective when it is undertaken by a leadership coalition that unites leaders of both (all) groups than when one or more group leaders do not take part in such a coalition” (Hogg et al., 2012: 245). Empirical studies examining the seven intergroup leadership theory propositions are still extremely limited (Kershaw et al., 2021a; Kershaw et al., 2021b; Rast et al., 2018; Rast, van Knippenberg, & Hogg, 2019; Porck & van Knippenberg, 2020) and highlight the compelling need for further study in this area.

Next, *what leader knowledge, skills, and abilities (KSAs) promote intergroup leadership and in turn effectiveness in a multiteam system?* As noted in the problem statement, little to no research exists examining KSAs promoting successful intergroup leadership and MTS performance. Adequately preparing an individual to lead a complex MTS may be more involved than simply providing a short course prior to taking charge of an MTS. Are there steps organizations can take to develop its workforce more deliberately through career broadening assignments that expose future leaders to all

aspects of the organization's operations and expand their interaction with and understanding of internal and external stakeholders? Research has shown leaders with broad functional experience are better equipped to understand the entire system and how individuals and teams operate within it successfully (Bunderson & Sutcliffe, 2002). Broad functional experience can also enhance horizontal coordination in an MTS but that breadth of experience may dampen aspirational behavior within the MTS (DeVries, Walter, van der Vegt, & Essens, 2014). This negative effect, however, can be mitigated through appropriate vertical coordination between component teams and leadership teams (DeVries et al., 2014).

In addition, successful MTS performance depends on the leader's ability to understand and communicate how each component team works together as part of the system to achieve the superordinate goal (Mathieu et al., 2018). Are leaders with higher levels of systems thinking skills better equipped to exercise the leadership required to drive MTS performance? While no empirical studies have considered the role played by systems thinking in effective intergroup leadership or successful MTS performance, research shows leaders who provide an overall vision and understanding of how the MTS operates and effectively communicate that to their teams are more successful (DeChurch & Marks, 2006). In addition, understanding system operations allowed leaders to better align and integrate component team actions toward superordinate goal accomplishment (Marks et al., 2005).

Two different but related aspects of identification are at play in an MTS. As noted above, intergroup leadership centers around the development of an intergroup relational identity through appropriate leader rhetoric and behaviors that demonstrate the

importance of maintaining a distinctive identity as an individual or team while also building an identity that extends beyond the self or team to include a sense of place in relation to others. A second aspect of identification relates to the focus of identification, either with the component team or the MTS. Given many MTSs are made up of component teams with diverse skill sets and may represent different organizations with different cultures, norms, and traditions, what role does the leader's foci of identification play in successfully leading a component team or the MTS? Empirical evidence thus far is mixed, with some studies suggesting identification with the MTS is more important for MTS processes and performance (Cuijpers, Uitdewilligen, & Guenter, 2016), strategic consensus (Porck, van Knippenberg, Tarakci, Ateş, & Groenen, 2020), and effective intergroup relations (Richter et al., 2006) while others argue MTS identification can increase uncertainty and deplete resources needed to drive MTS performance (Porck, Matta, Hollenbeck, Oh, Lanaj, & Lee, 2019) and that the effects of identification on MTS performance may not be linear (Mell, DeChurch, Leenders, & Contractor, 2020). A field study which captures the full extent of a leader's breadth and depth of experience and assesses their systems thinking skills and foci of identification offers a significant step forward in furthering our understanding of influences on effective intergroup leadership in an MTS.

While it is important to understand individual dynamics and leadership actions within an MTS, it is also important to consider influences on team functioning, both at the component team and MTS level, and subsequent effects on system performance. Thus, my final research question, *“how do team identification and functional diversity influence the relationship between intergroup leadership and effectiveness in a multiteam*

system?” seeks to address these essential elements of MTS effectiveness. How does a component team’s collective foci of identification influence leader actions? If a component team leader, for example, has higher identification with the MTS while his or her team exhibits stronger identification with the component team, does that help or hinder intergroup leadership and in turn MTS performance? Does this identity mismatch between leader and team demand higher levels of leader rhetoric to build a stronger intergroup relational identity in order to overcome a focus on subgroup self-interest? Existing research has typically examined identification with the component team and MTS as an aggregated measure of individual assessments (e.g., Cuijpers et al., 2016 and Porck et al., 2019). This study examines the strength of identification with the component team and MTS from both an individual leader and a team perspective.

Finally, in a similar fashion, does the relative homogeneity or heterogeneity of functional expertise resident in the MTS require higher levels of leader rhetoric to build the intergroup relational identity needed to move beyond subgroup self-interest and achieve effective levels of performance? Research has shown higher levels of homogeneity in component teams and MTSs can promote cooperation since members “speak the same language” and understand where to target their requests for information. High levels of heterogeneity can lead to conflict due to misunderstandings and communication gaps driving ingroup and outgroup divisions (Mathieu, 2012). The research setting for this study is a perfect venue to explore answers to these questions, with varying levels of functional diversity within squadrons, groups, and wings within the Department of the Air Force.

Research Setting

This study answers the repeated call to study “MTSs in the wild” (Burke, 2014; Shuffler et al., 2014) such as humanitarian assistance and disaster relief operations examined by DeChurch and colleagues (2011), Mathieu’s (2012) conceptual examination of a high school development MTS, and Shuffler and colleagues (2014) edited volume of case studies such as an examination of patient care as an MTS, coordinating mechanisms in MTSs within the petroleum industry, and MTS challenges experienced by NASA in spaceflight operations. It also is in line with the recommendation to move the level of analysis to the MTS (DeChurch, Hiller, Murase, Doty, & Salas, 2010; DeChurch & Zaccaro, 2010; DeChurch & Mathieu, 2009) due to complex motive/goal structures of multiple teams at multiple levels, interdependencies within, between, and across systems as well as significant variation in the size and distribution of component teams within the MTS.

The setting for this study is the complex, nested structure of wings, groups, and squadrons in the Department of the Air Force. Figure 1 depicts the typical hierarchical structure of these units. There are 11 Major Air Commands (MAJCOMs) or Direct Reporting Units (DRUs) that report to the Chief of Staff of the United States Air Force. Each MAJCOM is composed of two or more Numbered Air Forces (NAF) or Centers. In turn these entities include multiple subordinate units, which are typically wings but in some cases one or more groups. Each wing includes two or more groups, and each group includes multiple squadrons. In some cases, units not large enough to earn designation as a squadron are organized as a “numbered” flight.

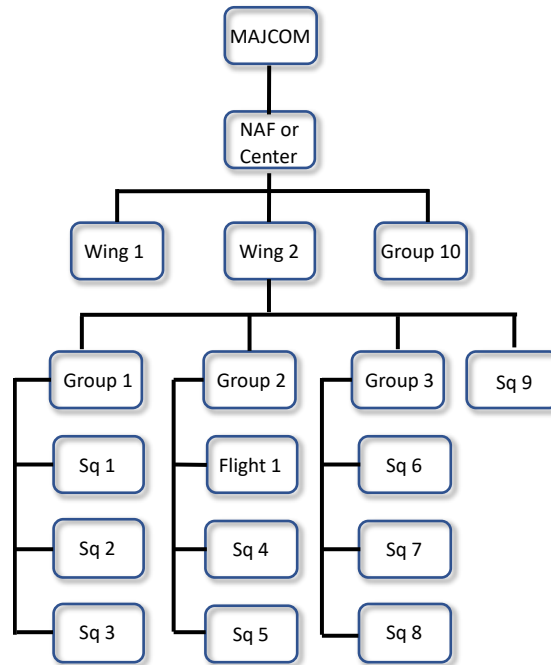


Figure 1. Hierarchical Structure of Air Force Units

To date no empirical studies have examined issues of leadership and effectiveness in such complex MTSs. In a conceptual paper DeCostanza and colleagues (2014) used the example of an Army Brigade Combat Team (BCT) to describe a “hierarchical MTS embedded in a parallel MTS.” They highlighted the high level of interdependence between component teams and the double-edged sword of geographic dispersion promoting effective operations and enhancing force protection while also creating communication and coordination challenges.

An Air Force wing experiences the same challenges as the BCT. Figure 2 illustrates a hierarchical MTS (such as a group) embedded in a wing (also an MTS)

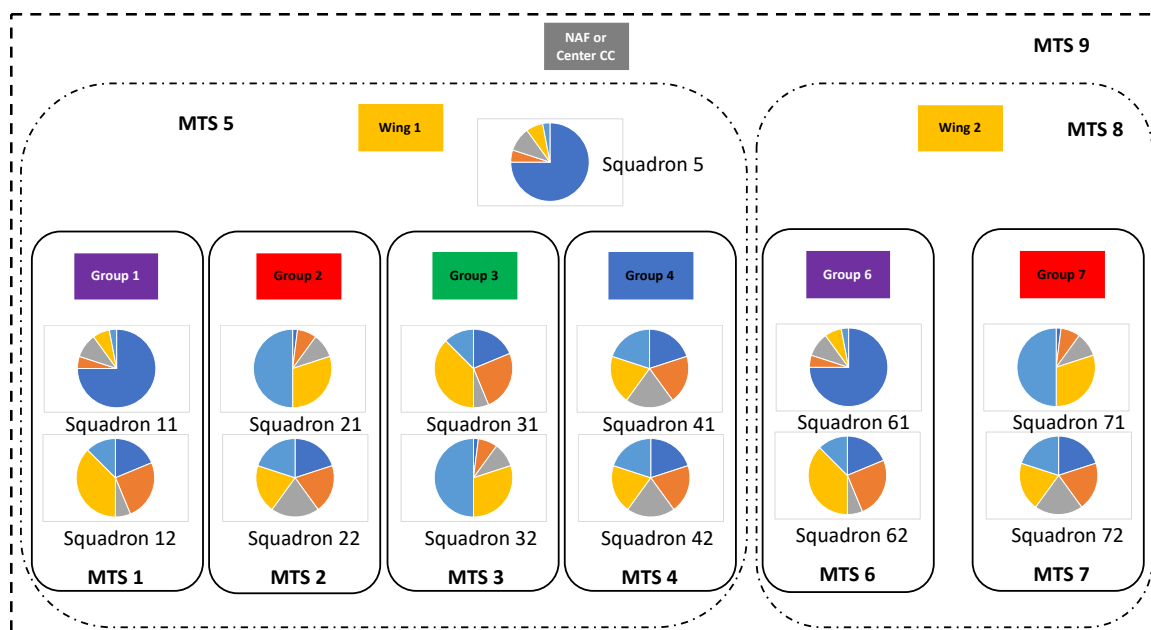


Figure 2. Example of Hierarchical and Nested MTS Structure

which in turn is part of a NAF or Center (also an MTS). A wing may include 2 or more groups often with very different functions (e.g., operations, maintenance, mission support, and medical) and range in size from under 1,000 to over 8,000 personnel. A group may include squadrons that are relatively homogenous in terms of their functional diversity (e.g., an operations group with two nearly identical flying squadrons and an operations support squadron) or exhibit high degrees of heterogeneity with each squadron having a different functional focus and within each squadron varying levels of diversity (e.g., a Mission Support Group composed of squadrons focused on security; communications; personnel, morale, and welfare; logistics activities such as supply and transportation; contracting and purchasing; and finally engineering activities to include facility maintenance, repair, and construction, fire and emergency services, emergency management, and environmental compliance, protection, and remediation). Groups

typically range in size from several hundred to as large as 3,000 or more individuals. Squadrons have at least 35 individuals and can be as large as 900 or more personnel.

In addition, wings that serve as the host unit for an installation are responsible for providing installation support services for all organizations assigned to that installation. For units providing these services (Mission Support Groups, Medical Groups, and wing staff agencies such as chaplains and legal services) this added complexity can range from less than 10 additional “mission partners” to large joint bases supporting several hundred units from all military branches, defense agencies, and other government agencies. The higher the number of mission partners the more important activities such as boundary spanning become and the more likely it is for these units to experience high levels of goal discordancy, having to balance the needs and superordinate goals of their parent wing while at the same time ensuring support provided to mission partners allows these units to succeed.

In line with DeCostanza and colleagues (2014), the following example illustrates the complex environment in which squadrons, groups, and wings operate on a daily basis. Generic Air Force Base (GAFB) is home to a fighter wing, an air refueling wing, and a rescue group. GAFB also supports an Army headquarters commanded by a general officer and a defense research agency. The fighter wing, as the designated host unit for GAFB, is responsible for all installation support activities (i.e., city services) such as facility maintenance, security, communications, medical, legal, and chaplain support and must deliver that support to all units assigned to GAFB. The fighter wing, air refueling wing, and rescue group each report to a different NAF and MAJCOM, resulting in five different reporting chains for the major organizations represented at GAFB.

An average day at GAFB might include the production of 20 fighter missions, five air refueling missions, a ceremony hosting multiple high-ranking distinguished visitors at the Army headquarters, and a significant test of a new prototype weapons system being conducted by the defense research agency. Within the fighter wing, the operations group ensures aircrews are trained and ready, the maintenance group provides mission-capable aircraft, the medical group ensures wing personnel are medically ready to perform their tasks while the mission support group ensures the airfield, facilities, logistical support, security, and communications are maintained and ready to support activities for the day. No sortie (one mission by a single aircraft) can be generated without each unit in the wing accomplishing its proximal goal in support of the ultimate wing superordinate goal to launch and recover the fighter missions planned for the day. At the same time, the medical and mission support groups are also supporting the missions and priorities of GAFB's mission partners requiring those group commanders to balance resource allocation (people, time, money) to ensure mission-essential needs across the installation are met.

Symbols of unit pride and *esprit de corps* are numerous. Unit patches, guidons, coins, and individual functional badges are just some of the many artifacts that reinforce an Airman's identity with their unit or functional specialty. Base-wide events such as award ceremonies and intramural sporting events can reinforce unit boundaries and promote ingroup and outgroup comparisons. Commanders at all levels seek to balance messaging that instills clarity of purpose within their unit while at the same time promoting a sense of place that lets their team know how they contribute to the larger mission of the wing and the installation.

Thus, the research setting provides a unique opportunity to examine concepts critical to intergroup leadership in an MTS – multiple foci of identification, the use of rhetoric to instill an intergroup relational identity, and varying levels of functional diversity within an MTS.

Conceptual Framework

This study seeks to understand what it takes to effectively lead a complex system of teams such as nested MTSs found in the Department of the Air Force (Figure 2) and understand the role leader experience, systems thinking, functional diversity, and social identification may play in promoting or inhibiting intergroup leadership and ultimately MTS effectiveness. The research model (Figure 3) explores five hypotheses focused on identifying drivers of intergroup leadership and MTS effectiveness using a sample of 256 commanders in 180 MTSs at the squadron, group, and wing level through a combination of archival data and survey responses from the commanders, their subordinates, and supervisors. The proposed research model (Figure 3) depicts the hypothesized relationships and highlights the level at which the study variables were analyzed.

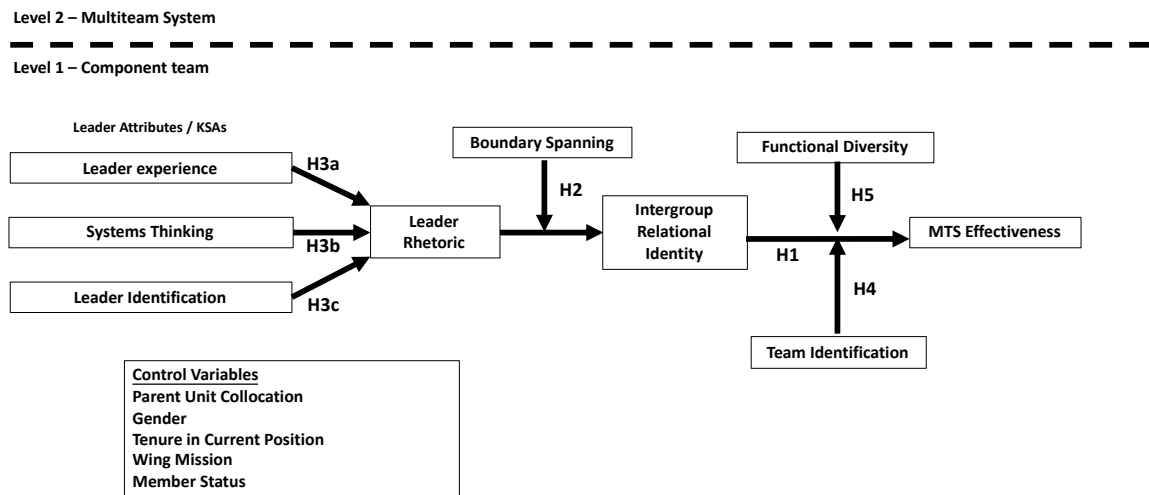


Figure 3. Research Model

Summary

In summary, this study breaks new ground in several areas surrounding the study of intergroup leadership and MTS effectiveness in research as well as practice. First, the study contributes to the growing body of knowledge regarding leadership in multiteam systems by conducting a field study examining complex MTSs, thereby filling a gap in understanding MTSs in a field setting (Burke, 2014; Davison et al., 2012; Shuffler et al., 2014). This study can also help researchers understand if measures of MTS attributes and effectiveness developed and supported in laboratory simulations translate to a field setting. Third, it is the first study, to my knowledge, to use the systems thinking scale to measure system thinking skills and apply the construct in the context of intergroup leadership in an MTS. This provides not only additional empirical validation of the systems thinking scale but also introduces this construct as an important consideration for future studies on intergroup leadership and MTS effectiveness. Finally, this study can begin a conversation in both research and practitioner communities surrounding intergroup leadership such that a consistent approach to developing it in leadership development programs and applying it in MTS settings can begin to take shape.

CHAPTER 2: LITERATURE REVIEW

Theoretical Background

This literature review is grounded in the MTS literature stream and will begin by reviewing the genesis of the MTS as an organizing construct for the study of team dynamics, the use of an MTS as a framework to understand the dynamic nature of these systems, and the characteristics that distinguish this collective from other team structures. Within the MTS framework, the review will focus on the importance of leadership (grounded in intergroup leadership theory) and identification as drivers of MTS performance. It will next introduce systems thinking as a critical KSA envisioned to act as an antecedent to intergroup leadership. The overview concludes with a discussion of the various ways in which MTS effectiveness has been conceptualized in theoretical work and measured in empirical studies. The review will also highlight significant recent and relevant conceptual and empirical studies with a special emphasis on works related to leadership, boundary spanning, and identification. Support for specific hypotheses is derived from pertinent research from intergroup leadership, systems thinking, social identification, and diversity literature.

MTS – A “team of teams” with complex relationships and leadership challenges. As noted in the introduction, organizational and societal demands are driving an increased reliance on collectives of teams, both internal to an organization and across organizational boundaries to address complex challenges and solve difficult problems.

This “team of teams,” formally defined as a multiteam system is:

“Two or more teams that interface directly and interdependently in response to environmental contingencies toward the accomplishment of collective goals. MTS boundaries are defined by virtue of the fact that all teams within the system, while

pursuing different proximal goals, share at least one common distal goal; and in so doing exhibit input, process, and outcome interdependence with at least one other team in the system.” (Mathieu et al., 2001: 290).

Five characteristics are key when considering whether a collective system is an MTS (Mathieu et al., 2001). First, two or more distinct teams have interdependent members and share a proximal goal. Second, an MTS is typically larger than a team but smaller than an organization though this is not always the case. Third, input, process, and outcome interdependence exist between a component team and at least one other component team in the system. Fourth, the MTS is an open system and its configuration drives context-dependent performance requirements. Finally, a common superordinate goal is shared by all component teams in the MTS (Mathieu et al., 2001).

Component teams, such as fire fighters and Emergency Medical Technicians (EMTs) responding to the scene of an accident, are the building blocks of an MTS. Each component team may have different proximal goals that at some time must be linked at a higher level for the MTS to exist (Marks et al., 2005). Functional interdependencies -- input, process, and outcome interdependence (Zaccaro, Marks, & DeChurch, 2012) link component teams to each other. Input interdependence involves the sharing of resources by component teams, process interdependence looks at the level of interaction required to achieve MTS goals and output interdependence assesses the level to which a component teams' outcomes depend on other teams in the MTS accomplishing their goals (Zaccaro et al., 2012).

Much of the early MTS research focused on MTS transition and action processes, with Marks and colleagues (2005) finding that cross-team processes were a more accurate predictor of MTS performance than within-team processes and were more important

when interdependence levels between teams were high versus low. DeChurch and Marks (2006) examined leadership actions and processes in MTSs and found system performance was not improved when leaders focused on building strong component teams, but leaders engaged in cross-team processes focused on bridging gaps and differences between teams did improve MTS performance.

Compositional attributes (e.g., number and size of component teams, boundary status, and organizational and functional diversity), linkage attributes (e.g., interdependence, hierarchy, and power distribution), and developmental attributes (e.g., the genesis of the MTS, tenure, and stage of development) can vary greatly between MTSs and highlight the importance of context when studying MTSs (Mathieu, 2012; Zaccaro et al., 2012). Small, functionally homogenous component teams may be less prone to coordination breakdowns and miscommunication when compared to large MTSs composed of multiple component teams (small to large) with high degrees of functional heterogeneity (Mathieu, 2012). Geographic dispersion of component teams can increase the difficulty of operations and coordination when elements of the MTS operate on different time schedules or are located in different time zones (Zaccaro et al., 2012). MTSs established as a one-time entity to respond to a natural disaster may react and respond differently when compared to MTSs with fixed membership established for the long-term, or MTSs that evolve over time with changing composition (Mathieu, 2012).

The small volume of MTS research and the lack of specific information surrounding research populations and methods can make it difficult to draw broader conclusions surrounding MTS characteristics and effectiveness. Waring (2019) examined 13 MTS studies noting differences in the sample populations and methods used

to address the research questions. Case studies, historiometric analyses, and field studies often lack details describing the number and size of component teams as well as MTS structures, making it difficult to extend the findings to other situations and contexts.

While nearly all MTS research envisions the MTS as an organizing construct or level of analysis to evaluate areas such as leadership and coordination in a team of teams, Luciano, DeChurch, and Mathieu (2018) recently advanced a meso-theory of system functioning centered on two structural features, differentiation and dynamism.

Differentiation measures the degree of difference and separation between component teams over time while *dynamism* considers system stability and variability over time. Different forms of MTSs will function differently and those systems with high degrees of differentiation and dynamism will be more susceptible to breaking from forces that drive teams apart and destabilize the overall system (Luciano et al., 2018).

Boundary enhancing forces, generated through differentiation, tend to reinforce differences between component teams and increase the salience of team membership. A heightened sense of identity with the component team has the potential to direct individual attention and effort away from the MTS and toward component team goals, risking development of an identity threat and an “us versus them” mentality (Connaughton, Williams, & Shuffler, 2012; Hogg et al, 2012). In addition, both the lack of information and lack of clarity surrounding component team goals and activities can increase the salience of team boundaries and hinder coordination and cooperation. Firth and colleagues (2015) examined how inconsistent and hazy definition of shared goals and focus areas between component teams can impact performance. Frame-of-reference training positively influenced MTS performance by improving coordination between

teams and higher levels of within-team coordination had a positive effect on the relationship between the training and coordination between teams (Firth et al., 2015).

Disruptive forces generated from dynamism lead to increased uncertainty and instability in the system. Changes in goal hierarchies, task uncertainty, and diversion of attention when component teams must balance MTS tasks against outside influences all have the potential to disrupt MTS performance. A recent study by Porck and colleagues (2019) examined the influence of depletion and task uncertainty on identification in an MTS. They argued and found uncertainty surrounding MTS norms and processes led to higher levels of depletion which in turn lowered MTS performance for those who more strongly identified with the MTS while the certainty that comes from the comfort of working within a familiar setting (component team) and more strongly identifying with it led to lower levels of depletion and higher MTS performance.

Changes to the MTS structure, environment, or task can also create disruptive forces. Davison and colleagues (2012) demonstrated that in MTSs with specialized component teams (e.g., operations, support) leader coordination only improved systems performance when those actions were focused on the component team most critical to goal accomplishment. For example, the importance of different component teams over the life of a critical incident response will vary due to the dynamic nature of the MTS. All effort initially may be focused on emergency responders as they work collectively to put out a fire and save lives. However, once the fire is out and immediate danger mitigated, other teams involved in the clean-up and restoration of normal operations take primacy. In the military this is often referred to as a “supported / supporting” relationship, with the supported commander being the focus of activity and supporting

units directing their efforts toward the supported unit's success. The supported commander may change during the course of an operation depending upon the phase of the operational plan being executed. Taken together, the meso-theory seeks to explain the interplay of individuals, teams, and systems and how these boundary-enhancing and disruptive forces impact emergent collective states at the component team and MTS levels, potentially increasing focus on component teams at the expense of the MTS and making collaboration more difficult (Luciano et al., 2018).

Team of teams leadership – a balancing act. Leading an MTS is a complex challenge as leaders shift their time and attention from the component team to the MTS (Mathieu et al., 2001), develop messaging that helps team members make sense of their environment and understand the big picture of system functioning (DeChurch et al., 2011), and attend to the daily tasks needed to direct component team activity and priorities. Leadership also serves as an integrating mechanism by facilitating coordination of activities across MTS subsystems, monitoring and managing relationships within and beyond the MTS, anticipating and reacting to environmental changes, and mitigating potential adverse impacts from conflicting demands (DeChurch & Marks, 2006; Zaccaro & DeChurch, 2012). Integrating the actions of multiple component teams with interdependent goals requires leaders to model and influence healthy collaboration within the system (Carter & DeChurch, 2014; Zaccaro et al., 2012; Mathieu et al., 2001; Pittinsky & Simon, 2007). It is, at its core, a balancing act requiring skills to integrate actions across teams within the system while simultaneously maintaining appropriate differentiation and distinctiveness between those teams (Carter, Ascencio, Trainer, DeChurch, Kanfer, & Zaccaro, 2019).

Wijnmaalen, Voordijk, and Rietjens (2018) studied a Dutch military construction team deployed to Afghanistan and a civilian construction team in California over several months. Their case study highlighted the impact component team dynamics can have on the development of shared mental models and communication. Disappointing results, negative events, and lack of trust led to increased conflict and disappointment which in turn made component team coordination and collaboration more difficult. In both cases, lack of effective leadership at the MTS level further exacerbated coordination difficulties and hindered team effectiveness.

Intergroup leadership (Hogg et al., 2012) promotes effective cooperation and collaboration across groups while at the same time minimizing the potential adverse impact of conflict and competition. This in a nutshell is the challenge facing MTS leaders -- balancing the promotion of healthy component team relationships and pursuit of team goals against building productive relationships and focusing on goal attainment at the MTS level (Pittinsky & Simon, 2007). Strategies to improve intergroup relations include promoting exchanges between component team members, minimizing conflict and competition through management of resources and team interdependencies, and understanding how to balance a shared identity against the need to maintain a sense of distinctiveness and belonging (Pittinsky & Simon, 2007). Kershaw and colleagues (2021a) found leaders from outside a subgroup whose messaging focused on building an intergroup relational identity successfully reduced ingroup bias, even among subgroups who strongly identified with their subgroup. In a related study Kershaw and colleagues (2021b) examined the role perceived identity distinctiveness threat played in evaluations of leader effectiveness when the leader promoted a collective, dual, or intergroup

relational identity. This situation closely mirrored the teams described in the earlier construction team case study (Wijnmaalen et al., 2018). Results showed when identity distinctiveness threats were high, leaders promoting an intergroup relational identity were viewed most positively followed by those promoting a dual identity. Leaders promoting a collective identity were rated the lowest.

Leadership actions promoting goal setting, planning, and training team members to understand MTS operations can help MTS leaders build an intergroup relational identity (Marks et al., 2005; Mathieu et al., 2018). In particular, training targeted at developing a sense of place, allowing component teams to understand the role each team plays in the MTS, and how they each work together to promote system success will help instill an intergroup relational identity throughout the MTS (Mathieu et al., 2018).

Cuijpers and colleagues (2016) noted an intergroup relational identity allowed members to retain their sense of distinctiveness and importance while the sense of belonging that comes from simultaneously feeling similar to but different from other teams also helped mitigate intergroup conflict.

My team, your team, or our team – the role of identification in an MTS. Social identification plays a very prominent role in an MTS due to the push and pull between component team and MTS interactions and intergroup relationships. Ashforth and Mael define social identification as “the perception of oneness with or belongingness to some human aggregate” (Ashforth & Mael, 1989: 21). Social identification within an MTS is likely to include multiple foci of identification (Connaughton et al., 2012). It is important to understand social identity and identification issues within an MTS if we want to better understand drivers behind component team interactions such as establishing and

maintaining relationships. Four salient identities may be in play for individuals at any one time: individual, work group, organizational and occupation (Scott, Corman, & Cheney, 1988). An identity will become more salient depending on the situation, context, and interaction with other team members (Hogg & Terry, 2000). Interactions with other MTS members and frequent work on system-level initiatives or goals may make the MTS identification more salient (Connaughton et al., 2012). Likewise, the relative importance of component team and MTS goals can impact system performance. Concentrating on component team goals at the expense of system objectives can adversely impact MTS effectiveness while too much focus at the MTS level may diminish effort toward component team goals, negatively impacting both team and system performance (Connaughton et al., 2012).

Leaders at both the component team and MTS levels need to strike a balance between building strong component teams and establishing strong bonds between those teams. Research on intergroup relations finds strong team boundaries lead to negative perceptions of those outside the team which can inhibit the flow of information and exchange of ideas (Tajfel, 1982). The distinct capabilities and specialized knowledge each component team brings to the table need to be leveraged and celebrated for their contribution to system success but not to the extent that the strength of the component team identification detracts from MTS effectiveness (Connaughton et al., 2012).

Strong component team identification may lead to ingroup bias which makes team boundaries more salient (Ashforth & Mael, 1989; Hogg et al., 2012) and subsequently hinders cooperation and coordination and negatively impacts system performance (Connaughton et al., 2012). DeChurch & Zaccaro (2010) as well as Luciano and

colleagues (2018) joined Connaughton's team (2012) in arguing MTSs where members' social identification is more aligned with the MTS rather than their component team will perform better than cases where component team identification is stronger than MTS identification. However, existing research is limited and inconclusive, with some empirical studies supporting this argument (Cuijpers et al., 2016; Wijnmaalen et al., 2018; Wijnmaalen et al., 2019) while Porck and colleagues (2019) found the opposite effect. Mell and colleagues (2019) found potential nonlinear effects of social identification in an MTS in an examination of how identity asymmetries affect interteam information sharing and performance. Clearly more empirical studies are needed in this area to understand the circumstances under which identification at the MTS level is most conducive to MTS performance and when component team identification may lead to better performance.

The whole is greater than the sum of its parts – the role of systems thinking. An MTS by definition is a complex system with multiple forces acting at times in opposite directions, making coordination within the system and leadership of it more challenging. DeChurch and Mathieu (2009) and Marks and colleagues (2005) highlight this difficulty by noting individual component teams can be very successful and satisfy their proximate goals while the overall MTS fails to achieve its distal goal(s). Collective success of the system depends on effective coordination and joint effort toward distal goal accomplishment. Efforts such as promoting too much component team empowerment and cohesion (Shuffler & Carter, 2018), goal discordancy, and other boundary enhancing forces (Luciano et al., 2018) can inhibit collaboration, disrupt system functioning, and ultimately hurt MTS performance (Carter, Cullen-Lester, Jones, Gerbasi, Chrobot-

Mason, & Nae, 2020; Marks et al., 2005). DeChurch & Zaccaro (2010) as well as DeChurch and colleagues (2011) examined several high-profile failures such as the response to Hurricane Katrina. They attributed failure in these cases to system level issues, noting systems fail more often due to between team issues than within team problems. Influences on or actions of one team in this interconnected system may ripple through other teams and have adverse impacts on overall system performance (Zaccaro & DeChurch, 2012).

One of the most essential functions of MTS leaders is providing an overall vision and understanding of how the entire MTS operates and the way in which component teams relate to each other and work in tandem to ensure system success (DeChurch & Marks, 2006). Leaders need to understand how the whole system works and individual teams contribute to success such that they can align and synchronize component team actions toward the collective goal (Marks et al., 2005). Ensuring team members make the connection between team and system goals and their impact on system performance is essential to success (Carter et al., 2019). Hogg (2015) noted effective intergroup leadership strategies mix rhetoric focused on promoting intergroup collaboration and cooperation and describing how this drives system success coupled with leadership action to model this behavior. Strategic communication to convey that “the whole is greater than the sum of its parts” and success of the system is possible because of the unique contributions and capabilities of component teams is critical to success.

Component team leaders play an important role in developing not only a mental model of how the team operates but also extending that shared cognition to the system level, so all members understand how teams work together to achieve MTS goals

(Murase, Carter, DeChurch, & Marks, 2014; Shuffler & Carter, 2018). Leaders must “think at the system level to not only provide support and task related behaviors but also understand how to manage individual components of the MTS” (Shuffler et al., 2014: 12). System knowledge helps leaders understand where to target their efforts at promoting integration, collaboration, and other between team activities such that their investment of time will yield the most benefit (DeChurch & Marks, 2006; Murase et al., 2014; Shuffler & Carter, 2018; Zaccaro & DeChurch, 2012).

MTS effectiveness – measuring team success. As originally conceptualized, MTS effectiveness is defined not by how well each component team meets its proximal goals but ultimately how component teams collectively meet goals at higher levels of the goal hierarchy (Mathieu et al., 2001). It is also context dependent. MTS effectiveness in a laboratory setting depends on the success metrics built into the simulation (Resick et al., 2012) and those measures may not be appropriate or translate to provide an accurate assessment of an MTS in a field setting, such as an emergency response MTS engaged in a live disaster exercise (Mathieu et al., 2018).

Research specifically explaining MTS effectiveness is limited (Mathieu et al., 2018). Individual component team success does not necessarily translate to overall MTS success as it depends on collective efforts at multiple levels. “No individual team can single-handedly accomplish an MTS superordinate goal” (DeChurch & Mathieu, 2009: 274). Luciano and colleagues (2018) note that system performance may suffer in situations where component teams have incompatible proximal goals or have their attention diverted by responsibilities outside their MTS.

The high degree of interdependence between component teams suggests MTS effectiveness depends on component team coordination (Davison & Hollenbeck, 2012). Marks and colleagues (2005) examined the impact of between- and within-team processes as a function of interdependence among component teams in an MTS. When the task was highly interdependent, successful performance was predicted by between-team processes but when team tasks were more independent, between-team processes were not a significant driver of MTS performance. The finding of between-team processes being more critical to MTS success was confirmed in additional studies (Davison et al., 2012; DeChurch & Marks, 2006). Interdependence between component teams also places a premium on high quality coordination and collaboration. MTSs with effective coordination and collaboration enjoy better component team and system performance (Davison et al., 2012; DeChurch & Marks, 2006; DeChurch & Zaccaro, 2010; Marks et al., 2005; Mathieu et al., 2001). System size, task complexity, and high levels of diversity with respect to skill levels and functional specialties can make effective coordination and collaboration difficult to achieve (Davison et al., 2012; Shuffler et al., 2015). When considering the interdependence of goal hierarchies, Marks and colleagues (2005) found that the more interdependent the goal hierarchy the more MTS effectiveness depends on effective action processes at the MTS level. In cases of less interdependence effective component team processes were the primary driver of MTS effectiveness.

Carter and colleagues (2019) argue MTS effectiveness is driven through border processes, specifically boundary spanning and interteam leadership. In addition, they concur with DeChurch & Marks' (2006) finding that effective MTSs have leaders that

can shift between strategy and coordination activities within and between teams. Their study empirically verified that training leaders to understand system operations increased leadership and coordination and leadership was positively related to MTS performance (DeChurch & Marks, 2006). Davison and colleagues (2012) found a more specific effect, with leadership actions focused on the component team most integral to the task generating the biggest positive impact on system performance.

Frame-of-reference training or understanding how parts of the system relate to each other, helped minimize representational gaps in component team member understanding of the MTS task leading to improved coordination between teams and in turn improved MTS performance (Firth et al., 2015). This requirement to develop a system-level understanding of component team interdependencies and MTS goal hierarchies highlights the importance of a leader's systems thinking skills and their ability to develop strategic messaging which defines an intergroup identity as an intergroup collaborative relationship (Hogg, 2015; Hogg et al., 2012).

In summary, significant gaps in knowledge remain when considering how to lead a team of teams, understand what factors may act as antecedents to effective intergroup leadership, and what influence team identification and diversity may have on the relationship between intergroup leadership and MTS effectiveness. In the following sections I develop specific hypotheses to address these questions, beginning with a discussion of intergroup leadership which is based on the role leader rhetoric plays in the development of an intergroup relational identity and how leader boundary spanning actions can enhance the effectiveness of this rhetoric as the leader "walks the talk." The discussion then turns to an examination of leader knowledge, skills, and abilities that may

act as antecedents to intergroup leadership and concludes with a look at the predicted moderating role of team identification and team functional diversity on the relationship between intergroup leadership and MTS effectiveness.

Hypothesis development

Leader rhetoric, intergroup relational identity, and boundary spanning. Team cohesion and strong team identity are generally heralded as worthy goals for team leaders and managers. Yet this focus on increasing cohesion often inhibits collaboration outside the team by introducing ingroup and outgroup dynamics which make it challenging for teams to act in the system's best interest rather than in the best interest of their team (Carter et al., 2019; Mathieu et al., 2001). Intergroup relations research is extensive, examining how to reduce conflict and prejudice but its focus is primarily how to change intergroup perceptions and behavior, offering little on how leadership influences intergroup relations (Hogg, 2015).

Hogg and colleagues' (2012) intergroup leadership theory logically extends concepts from intrapersonal relations to an interpersonal level and intragroup relations to an intergroup level. For example, the concept of an interpersonal relational identity between a parent and child can be translated to a group level where the relationship between students and teachers or nurses and doctors can form an intergroup relational identity (Hogg et al., 2012). Identification at a team level, whether developed as a collective or relational identity, extends an individual's self-identification to include others. While a collective identity focuses on similarities between an individual and others in the MTS, a relational identity relies on an individual understanding how they relate to others (Hogg et al., 2012). Herein lies a challenge for intergroup leadership –

when should leaders build a collective identity and when is building a relational identity a more effective strategy?

Intergroup relations are often fraught with identity clashes between different groups. Functional experts may view others outside their group as less important to goal accomplishment and in a military setting, unit pride and *esprit de corps* can trigger rivalries that undermine working relationships and overall team success. Intergroup leadership then must carefully navigate the various social identities at play in group members and across the MTS and avoid provoking threats to members' distinctive identities (Hogg et al., 2012; Rast et al., 2018). Rast and colleagues (2018) conducted three studies to evaluate whether a collective identity or relational identity was more effective in diffusing the adverse impacts of identity threats to leader performance. Results showed when members perceived an identity threat, leaders promoting an intergroup relational identity received better evaluations and were perceived as more effective than leaders promoting a collective identity. In the absence of such a threat, leaders promoting a collective identity were rated as more successful (Rast et al., 2018).

Two recent studies further examined the how perceived identity distinctiveness threats can influence the effectiveness of collective, dual, or relational identities developed by leaders in their teams. Kershaw and colleagues (2021a) first examined the interaction of identity threat with perceptions of the leader as either an ingroup or outgroup member (e.g., one of us or one of them). In two studies they examined whether being perceived as an ingroup or outgroup leader was more effective with differing levels of identity threat. They found leaders viewed as "one of them" were successful in lowering ingroup bias among members with high levels of identification when they

espoused a relational identity as compared to a collective identity but ingroup bias increased for those with weaker levels of identification. For leaders considered “one of us” the type of identity the leader advocated had no impact on ingroup bias. Their second study examined evaluations of leadership effectiveness when leaders advocated collective, dual, or relational identities under low and high levels of identity threat (Kershaw et al., 2021b). Diminishing a group’s distinctive identity risks interfering with an individual’s social identity and increases the perception that the group’s distinctive boundaries are being eroded (Kershaw et al., 2021b). Building a dual identity as a strategy can be beneficial since “the superordinate identity is salient but conjoins with a salient subgroup identity (a ‘different group working together on the same team’ representation)” (Dovidio, Gaertner, & Lamoreaux, 2009: 9). Results supported the hypothesis that threats to identity would show leaders supporting a relational identity were viewed as most effective followed by dual identities and finally collective identities.

Hogg and colleagues (2012) argue effective intergroup leadership builds a social identity with the group but also informs their understanding of that identity in a way that moves the group toward successful performance. “Through intergroup leadership rhetoric that provides a shift from a more instrumental intergroup relationship to an intergroup relationship that is part of group identity, members will, over time, come to view the nature of the relationship as identity defining, and they will be more likely to internalize it, through the process of self-categorization, as part of an intergroup relational identity” (Hogg et al., 2012: 242). Thus,

Hypothesis 1: Leader rhetoric has a positive indirect relationship with MTS effectiveness through intergroup relational identity.

Developing an intergroup relational identity through appropriate rhetoric is only one part of the equation. Team members not only listen to what leaders say, they watch what they do and conflict between leader rhetoric and action can adversely affect the development of a relational identity and team performance. Leader actions need to reinforce and model appropriate intergroup collaboration (Hogg et al., 2012). Team leaders often function as boundary spanners who have established links and relationships with other groups internal and external to the MTS. These connections can help defuse potential clashes and promote healthy working relationship between teams (Ancona & Caldwell, 1990; Hogg et al., 2012). Research has shown boundary spanning leadership is critical to successful intergroup leadership and intergroup relations (Ernst & Yip, 2009; Richter et al., 2006) and essential to building and sustaining an intergroup relational identity because it enables the leader to embody their rhetoric focused on building that identity and maintaining productive relationships with other groups (Hogg, 2015).

Porck and van Knippenberg (2020) in a recent study of middle manager boundary spanning argued it was important for managers to understand the conditions under which boundary spanning leadership will generate the most benefit to team success. They predicted boundary spanning activities would be most important for interdependent teams residing in different divisions due to the unique culture, language, and norms that often define different teams. Findings showed effective boundary spanning by middle managers improved intergroup effectiveness and should be viewed as a critical activity in building relationships across teams (Porck & van Knippenberg, 2020).

Therefore, boundary spanning serves as an important mechanism for leaders to model and demonstrate the importance of intergroup collaboration and coordination to

overall system success. Modeling this behavior, or “walking the talk” provides team members with a positive example to follow and in turn makes the impact of the leader’s advocacy for an intergroup relational identity more effective. Thus, in line with Hogg and colleagues (2012) I posit:

Hypothesis 2: Boundary spanning leadership moderates the positive relationship between leader rhetoric and intergroup relational identity such that intergroup relational identity increases as boundary spanning leadership increases.

Antecedents to intergroup leadership – leader experience, systems thinking, and identification. Identifying and empirically testing leader KSAs as antecedents to intergroup leadership will help organizations modify existing or implement new leader training and development programs that promote an understanding of leading within complex systems and in turn equip leaders to exercise intergroup leadership to support system effectiveness.

Leader experience. Research has demonstrated the link between effective interteam coordination and the breadth of functional experience present in team members (Bunderson, 2003). Leaders with higher levels of experience outside their primary functional area are better able to help team members understand what other functional areas contribute to the team and how they fit within the broader system and organization (Bunderson & Sutcliffe, 2002). DeVries and Colleagues (2016) found horizontal coordination between component teams was improved when members had a more general background (high intrapersonal functional diversity, IFD) but that high IFD may also dampen enthusiasm for tackling more challenging and specialized tasks. Effective

vertical coordination across hierarchical levels in the MTS helped mitigate IFD's potential negative effects on MTS performance.

MTS success depends on the extent to which its members share a common understanding of the MTS purpose and share a mental model of what superordinate goal accomplishment looks like (Mathieu, 2012). Individuals with broader functional backgrounds have more individual cognitive capacity for making sense of and understanding a variety of social situations. This in turn helps them deal with dynamic team situations and increase the likelihood they will engage in more boundary spanning actions that lead to successful outcomes (DeVries et al., 2014).

Work assignments designed to stretch leaders and develop skills needed to deal with solving novel and poorly defined problems are critical to leader development (Mumford, Marks, Connelly, Zaccaro, & Reiter-Palmon, 2000a). These same leaders must be able to develop solutions that address the complex environments in which organizations operate today, placing a premium on their ability to develop "organizational wisdom" and understanding of system-level operations (Mumford et al., 2000a). Job relevant knowledge developed through work experience and understanding of functional responsibilities provides the foundation needed to develop solutions to complex problems and drive successful leader performance (Mumford, Zaccaro, Harding, Jacobs, & Fleishman, 2000b).

Therefore, leaders with more diverse backgrounds are more likely to be able to develop strategic communication and rhetoric that builds an intergroup relational identity within their team. Further, the leader's varied background and experience will improve the quality of their boundary spanning activities since they are better equipped to know

where to target those activities for maximum benefit. Taken together, this leads to the following hypothesized relationship:

Hypothesis 3a: The indirect effect of leader attributes and KSAs on the development of an intergroup relational identity through leader rhetoric is moderated by boundary spanning leadership (intergroup leadership) such that higher levels of leader experience diversity will be more positively related to the development of an intergroup relational identity.

Systems thinking. The broader a leader's experience the more likely it is they will develop an appreciation of what it takes to make various aspects of a system or team of teams work in harmony (Bunderson & Sutcliffe, 2002). An MTS is a system and as such it is important for leaders at the MTS and component team levels to have systems thinking skills which enable them to develop a holistic point of view and understanding of the MTS. As Senge (1994) notes in his seminal work, *The Fifth Discipline*, the foundation of a systems thinking approach is an ability to identify and focus on interrelationships rather than looking at the individual components in a linear fashion. It is "seeing processes of change rather than snapshots" (Senge, 1994: 73). Systems thinking is defined as "a set of synergistic analytic skills used to improve the capability of identifying and understanding systems, predicting their behaviors, and devising modifications to them in order to produce desired effects. These skills work together as a system" (Arnold & Wade, 2015: 675).

Leaders should strive to communicate the overall MTS goal in a way that allows each component team to understand and appreciate how they contribute to accomplishment of this superordinate goal and how their actions support other

component teams (Mathieu et al., 2018). This clarity of purpose can help team members understand their place in the system and minimize the potential team members focus on distractions that may detract from goal accomplishment (Davis & Casey, 2018). A leader's ability to accurately convert their multiteam-interaction mental model to effective strategic communication with the team increases the likelihood team members will develop more accurate mental models and improve between-team coordination (Murase et al., 2014).

Therefore, the leader's system thinking capabilities are essential to the development of an understanding of system operations, how individual component teams bring their expertise to bear on system performance, and to the development of rhetoric the leader can use to share this mental model with his or her team. Taken together, leaders are able to construct an intergroup relational identity and model that behavior through boundary spanning actions appropriately targeted to the right audience at the right time based on their system knowledge. Thus, I predict:

Hypothesis 3b: The indirect effect of leader attributes and KSAs on the development of an intergroup relational identity through leader rhetoric is moderated by boundary spanning leadership (intergroup leadership) such that higher levels of systems thinking skills will be more positively related to the development of an intergroup relational identity.

Leader identification as an antecedent. Identification is envisioned to play two roles, first as an antecedent to intergroup leadership and second as a modifier of the relationship between intergroup leadership and MTS effectiveness. Given the research is

inconclusive regarding the effect of identification on MTS performance, examining its potential relationship with two different constructs can add to the conversation.

It can be difficult for MTS members to navigate the multiple identities operating at any given time in an MTS – individual, component team, and MTS (Carter et al., 2019; Hogg et al., 2012; Shuffler & Carter, 2018). Conceptual papers by DeChurch and Zaccaro (2010) and Connaughton and colleagues (2012) argue MTSs where members more closely identify with the MTS rather than their component team will perform better than MTSs whose members have higher identification with their component team.

Boundary issues can generate both interpersonal conflicts and goal conflicts. Component teams focused on their own goals at the expense of supporting MTS goals will result in lower levels of MTS performance (Connaughton et al., 2012). Likewise, if members believe their goals conflict with another component team cooperation can suffer and team members may limit the amount of time and energy they expend on system goals (DeChurch & Zaccaro, 2010). Finally, efforts to build strong cohesion and identification within a component team may aid in meeting component team objectives but also drive increased competition and conflict with other component teams and negatively impact MTS performance (Lanaj et al., 2013; Shuffler & Carter, 2018).

Several empirical studies evaluated the effect of component team and MTS identification on system performance with mixed results. In general, most empirical studies support the propositions developed in conceptual papers, namely that MTS identification has a more positive impact on MTS performance than identification with the component team. The relationship between work group identification in individuals performing boundary spanning activities and effective intergroup relations was positive

with high levels of MTS identification but not at lower levels (Richter et al., 2006).

Cuijpers and colleagues (2016) found system identification was more important for MTS processes and performance than component team identification. Organizational identification supported intergroup strategic consensus, but team identification can have the opposite effect (Porck et al., 2020).

Two recent studies, however, predicted and found different results. Porck and colleagues (2019) examined the role of depletion and task complexity on social identification in MTSs. They argued MTS identification can increase uncertainty because goals, objectives, and norms, for example, are more abstract and less concrete at the MTS level. That uncertainty overly depletes cognitive resources in component team members and as such, identification with the MTS will negatively impact performance while component team identification (due to less uncertainty) will have a more positive impact. Results supported the prediction, and the effects were heightened when task complexity was high and muted when complexity was low. Mell and colleagues (2020) conducted a series of three studies to examine how identity differences affect interteam information sharing and performance. In other words, do foci of identity differences among component teams within an MTS affect the volume of information shared between teams and influence performance outcomes? The first study demonstrated that MTS-focused component teams (CT) shared information equally while CT-focused component teams shared less with MTS-focused teams versus CT-focused teams. In addition, the level of information sharing between teams positively predicted overall system performance. Study 2 hypothesized a difference in the motives for collaboration between CT-focused teams and MTS-focused teams with respect to collaboration. They

argued and supported the hypothesis that CT-focused teams would view collaboration on a transactional basis, seeing cooperation as “a means to an end and conditional on its instrumental value” (Mell et al., 2020: 1565). MTS-focused teams, on the other hand, would view collaboration in a more altruistic sense in that it is an objective in and of itself and not a conditional arrangement. Study 3 provided the most surprising finding in that additional analysis of findings in the first study showed potential non-linear effects of identification on MTS performance.

While the results are less than conclusive, we can state that behaviors and outcomes are different depending on the foci of identification. Whether that difference has a positive or negative effect on MTS performance requires additional research to both replicate previous studies as well as examine identification in different MTS contexts. Given the leader behavior in question is the utilization of leader rhetoric to develop an intergroup relational identity reinforced by boundary spanning actions, I argue the salience of intergroup relationships will make it easier for leaders who more strongly identify with the MTS to convincingly build and sustain this intergroup relational identity and therefore,

Hypothesis 3c: The indirect effect of leader attributes and KSAs on the development of an intergroup relational identity through leader rhetoric is moderated by boundary spanning leadership (intergroup leadership) such that leaders who more strongly identify with the MTS increase the positivity of the development of an intergroup relational identity.

Team identification as a modifier. Building on the previous discussion, most empirical studies to date predict MTS identification will have a more positive impact on

system performance than identification with the component team. Strong component team identification can make team boundaries more salient and increase threats to identity distinctiveness, making the development of an intergroup relational identity even more important for avoiding intergroup conflict and promoting system effectiveness (Hogg et al., 2012; Kershaw et al., 2021b; Rast et al., 2018). Component teams that more strongly identify with the MTS are less likely to resist and more readily accept leader messaging focused on building an intergroup relational identity, accentuating and enhancing the influence of intergroup leadership on overall system effectiveness.

Therefore, I predict:

Hypothesis 4: The indirect effect of leader rhetoric on MTS effectiveness through intergroup relational identity is moderated by team identification such that the relationship is stronger with higher levels of MTS identification than component team identification.

Functional diversity in teams. Diversity, defined as “differences between individuals on any attribute that may lead to the perception that another person is different from self” (van Knippenberg, De Dreu, & Homan, 2004: 1008), can generate effects on individuals or teams whenever people use a particular attribute to differentiate one person from another (Williams & O’Reilly, 1998). Functional diversity of team members can present a number of challenges for smooth and effective MTS operations and performance. Higher degrees of homogeneity promote higher degrees of cooperation and enhanced coordination in an MTS whose members are functionally similar since they are more likely to better understand each other, operate in similar patterns over time, and know where knowledge resides in the MTS (Mathieu, 2012). MTSs with higher levels of

heterogeneity are more likely to experience misunderstandings and communication breakdowns and are more susceptible to the development of ingroup and outgroup classifications that can reinforce differences and lead to conflict (Mathieu, 2012). Heterogeneity of team members can also increase conflict depending on the level of diversity and make the creation of a shared or intergroup identity more difficult (Connaughton et al., 2012). External MTSs with members from different organizations are likely to have different cultures, different norms, and potentially different languages, both spoken (e.g., English, Spanish, German) and jargon (e.g., technical terms, acronyms) (Connaughton et al., 2012; Mathieu et al., 2018) making coordination and cooperation more difficult.

Functionally diverse MTSs can be both beneficial for system performance and detrimental as well. Diverse membership and team diversity can be viewed positively since team members would have access to a broader range of knowledge, skills, background, and resources and this in turn leads to better performance (van Knippenberg & Schippers, 2007; Zaccaro et al., 2012). Zaccaro and colleagues (2012) note that functional diversity in MTSs may mirror findings from a study of top management teams (Greening & Johnson, 1997) which found functional diversity helped senior leaders perform better in crisis situations up to a point beyond which high levels of diversity negated the performance benefits. High degrees of diversity can also create difficulties in coordination as well as psychological divides (Shuffler & Carter, 2018). Luciano and colleagues (2018) note that high levels of differentiation in an MTS can increase the salience of team boundaries and team identities and in turn inhibit the cooperation and collaboration that required to build trust.

Leaders of MTSs with functionally diverse membership need to be aware of the influence component team and MTS identification may play on system functioning.

DeVries and colleagues (2014) found coordination between component teams in functionally diverse MTSs only increased when individuals were inspired to share what they knew for the good of the MTS while those without this focus regarded interteam coordination as a detractor and avoided the behavior.

Thus, functional diversity can be a force multiplier, or it can negatively affect system performance. The degree of its impact on MTS effectiveness is influenced by the leader's ability to communicate and build an intergroup relational identity such that team members understand how to harness and leverage that diversity to improve system processes and ultimately performance outcomes. Higher degrees of heterogeneity make this intergroup leadership challenge more difficult and therefore, I posit:

Hypothesis 5. The indirect effect of leader rhetoric that defines group identity in terms of a complementary intergroup collaborative relationship through the development of an intergroup relational identity on MTS effectiveness is stronger with higher levels of functional diversity.

CHAPTER 3: RESEARCH METHODOLOGY

Research Design and Rationale

The purpose of this study is to examine intergroup leadership in the context of a complex MTS, evaluate the influence of specific leader knowledge, skills, and abilities that may promote intergroup leadership and MTS effectiveness, and understand how functional diversity and social identification at the component team and MTS levels moderate the relationship between intergroup leadership and MTS effectiveness. The conceptual model described in Chapter 1 is a multi-level moderated mediation model, with both first stage and second stage moderators in addition to several leader attribute antecedents hypothesized to influence the rhetoric leaders use to build an intergroup relational identity and impact MTS effectiveness. The dependent variable for the full model is MTS effectiveness and intergroup relational identity serves as a dependent variable in Hypotheses 2 and 3a-3c.

Leveraging both MTS and intergroup leadership theory literature streams, this study examined intergroup leadership in MTSs using a study population of squadrons, groups, and wings in the Department of the Air Force. The study incorporated data collected from multiple sources. Archival demographic data was obtained from personnel, manpower, and real property databases. Individual responses to an online survey were obtained from supervisors, commanders, and subordinates. The survey was constructed with validated scales for leader rhetoric, intergroup relational identity, boundary spanning, systems thinking, and identification. Leader rhetoric and intergroup relational identity measures relied on subordinates to evaluate their supervisors. In addition, MTS effectiveness relied on supervisors to rate subordinate team performance

by using existing Air Force standards of performance. This approach minimized the risk of common method and single instrument bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003; Podsakoff, MacKenzie, & Podsakoff, 2011).

Sample and Procedure

The Department of the Air Force via the Air Education and Training Command (AETC) Transformational Education and Training Applications (ATETA) sponsored this research through an Educational Partnership Agreement (EPA) with the Drexel University LeBow College of Business. This no-cost arrangement allowed me as the student researcher and my dissertation co-chairs acting as principal investigators to gain access to archival data and conduct a survey of Department of the Air Force personnel. I held multiple discussions with key Air Force leaders, my dissertation co-chairs, and the chair of Drexel University's Institutional Review Board (IRB) prior to research protocol completion and submission to ensure it addressed appropriate safeguards for respondent information privacy and dataset security. The Drexel University IRB approved protocol number 2004007796-002 on 14 Jul 2020, the Air Force Human Research Protection Official (HRPO) approved the protocol on 17 Jul 2020, and the Air Force Survey Office (AFSO) approved survey distribution through Survey Control Number (SCN) AF20-090A1 on 22 Jun 2020.

This quantitative study used archival data obtained from the Air Force as well as survey data collected via the Qualtrics survey platform. The survey invitation and survey measures are located in Appendix A and Appendix B, respectively. The Air Force provided archival demographic data (Appendix C) from the Military Personnel Data System (MilPDS), the Defense Civilian Personnel Data System (DCPDS) and the

Manpower Programming and Execution System (MPES). An analyst in the Directorate of Manpower, Personnel, and Services, Headquarters United States Air Force at the Pentagon, Washington D.C. collected and deidentified, as appropriate, requested data elements from these systems. In addition, the Air Force Civil Engineer Center, Joint Base San Antonio (JBSA), TX provided real property data about installations (Appendix C) where the wings, groups, and squadrons reside. This information included the number of buildings and structures, total square footage maintained, total acreage, and the host unit for each Department of the Air Force installation. I did not collect similar data for units assigned to installations run by other services (e.g., Army or Navy) or entities as it was outside the research protocol scope.

I designed the survey instrument to balance ease of use with perceptions of privacy, using prepopulated and personalized items to minimize workload on the respondent while guarding against the possible perception of reduced anonymity by allowing participants to use a series of drop-down menus to select their unit of assignment. In addition, respondents selected the level of their immediate commander to direct them to the appropriate set of identification questions given some units skip an echelon of command in their reporting hierarchy. I tested the survey with a group of six retired Air Force officers ranging in rank from Lieutenant Colonel to Lieutenant General who had previously commanded units at the squadron through NAF or Center level. I made minor wording changes to improve clarity and consistency based on their observations and recommendations. I also adjusted the rating scale for MTS effectiveness to conform with standard operating practices within the Air Force.

The Air Force analyst provided a list of all possible units to which an Air Force member could be assigned. This “Personnel Accounting System (PAS) tree,” as of 31 Mar 2020, contained 15,605 potential units for inclusion in the survey population. I removed units outside the hierarchy of interest (squadron – group – wing), units assigned to the Air Force Reserve or Air National Guard, units whose identity is “data masked” and joint organizations to create an initial list of units (2,360) in the survey population. I next reviewed the hierarchy information in this list to check for errors or missing information and made corrections as appropriate. An updated PAS tree is generated each month to reflect unit activations, deactivations, and changes in reporting structure. I repeated this process with the PAS tree as of 31 May 2020, setting the initial survey population baseline at 2,314 units.

Using this population baseline, the Air Force analyst ran queries in MilPDS and DCPDS to identify individuals serving in leadership positions for the units outlined above and obtained their official Air Force e-mail address. I next built a survey contact list to distribute the survey invitation. We identified several issues at this time. First, individuals assigned as flight commanders within a squadron could not be precisely identified through the personnel systems. The original survey design targeted those serving in flight commander positions and unfortunately this was not possible. Second, there were individuals at the group level and above who shared the same Air Force Specialty Code (AFSC), i.e., the commander and deputy commander, making it difficult at this point to precisely identify which individual to target with the survey invitation. Finally, some of the units identified had no commander assigned in the personnel systems and some individuals did not have an official email address associated with their

personnel record. Therefore, we elected to over-sample the targeted population and revised the survey to include two screening questions intended to limit participation of those outside the targeted population.

Individuals in the survey population were predominantly members of the military but some support units led by Department of the Air Force contractor personnel could potentially be part of the survey population. Per Department of Defense guidance, contractor personnel are not eligible to participate in surveys. These individuals were identified and removed from the survey population prior to survey administration. In addition, some Department of the Air Force units include personnel from the Army, Navy, or Marine Corps. These individuals were ineligible to participate in the study as they are outside the jurisdiction of the Department of the Air Force. Finally, installations located outside the United States may have host-nation civilian personnel serving at the squadron level or above. These non-US citizens were also excluded from the survey population as we did not make a formal request to their host government to approve their participation.

The final survey contact list included 18,713 Department of the Air Force individuals who, as of 31 May 2020: 1) occupied active-duty numbered flight, squadron, group, and wing commander positions; 2) supervised wing or direct-report group commanders as a NAF/Center-level commander or higher; or 3) were supervised by the probable commander of a unit. Supervisors of wing/group commanders assessed performance of their subordinate wings/groups but did not complete the remaining portions of the survey instrument. In addition, due to the inability to identify with one hundred percent accuracy the current unit commander, a small sample of additional

personnel assigned to units at the squadron, group, and wing level were invited to participate. Two screening questions added to the survey limited participation of those outside the target population by ending the survey if the individual was not assigned as a commander at the group level or above and was not assigned to a squadron. There were 856 individuals who responded “yes” to these two screening questions and the survey ended at that point for these individuals.

The e-mail invitation to participate (Appendix A) included a transmittal memo from the Director, Directorate, Manpower, Personnel, and Services, Headquarters United States Air Force, and the survey link. In line with recommendations of Podsakoff and colleagues (2012), this short statement, in general terms, addressed the survey’s purpose, invited voluntary participation, and included a personalized survey link. The invitation also emphasized the measures being taken to ensure their anonymity, addressed how data would be deidentified prior to analysis, and stressed survey results would be reported in aggregate form only (Podsakoff et al., 2003).

Qualtrics limits the number of cells in contact lists when using drop down menus. This limitation required me to break up the single contact list with 18,713 names to 68 contact lists containing no more than 344 names per list. Each contact record contained 1) information about the recipient’s unit hierarchy, 2) names of subordinate units to evaluate for MTS effectiveness and 3) identification of the appropriate component team, MTS, superordinate group, and command level in addition to first name, last name, and e-mail address. The unit specific information was used as piped text in the survey to personalize the questions for each individual’s unit of assignment and command chain.

Survey administration began on 17 Aug 2020 and terminated on 30 Sep 2020 for nearly all survey participants. Surveys for supervisors of wing and group commanders remained open until 15 Oct 2020 to provide maximum opportunity for their voluntary assessment of subordinate unit performance. During survey administration several individuals expressed concern about survey invitation legitimacy and viewed it as a “phishing” attempt since the invitation originated outside the “.mil” e-mail domain. A clarifying e-mail was sent to all participants (Appendix A) and reminder e-mails sent once a week while the survey was active. In addition, several who received the invitation and attempted to take the survey noted issues with the drop-down menus from which they selected responses appropriate for their unit hierarchy. I built new contact lists and sent new survey links to these nine individuals and this action resolved the error, bringing the total number of survey invitations to 18,722. Other individuals, after receiving the clarification e-mail referenced above, requested to have their survey reset as they did not answer the screening questions correctly. I honored these requests, reset their survey, and sent a new link to allow survey participation.

Once the survey window closed, I downloaded the survey response files to my secure Drexel One Drive server and shared an access link with the Air Force analyst. The analyst used SAS to merge survey data with the requested personnel data from MilPDS and DCPDS. This resulted in a total of 18,852 records in a merged dataset which was uploaded to the secure One Drive server.

I noted several errors when reviewing the merged dataset. There were duplicate records (13), personnel data from individuals in units not part of the survey population (8) and records which upon merging the survey data and personnel data contained

identical survey information (109). I removed duplicate records with no survey response, records outside the survey population and records with errors on merge, leaving 18,722 records for analysis.

There were 6,238 responses to the survey with 5,208 complete responses (27.8%) and 1,030 partial responses (5.5%) for a total response rate of 33.3%. The research model focused on leaders at the component team and MTS level and therefore I needed to identify those in command positions. In several instances a unit had more than one individual with a commander duty title and/or AFSC or there was no commander included in the dataset. I coded individuals with a commander AFSC and duty title in one of 5 ways: current commander as of 31 May 2020 (1), former commander (2) (i.e., there were two commanders of one unit and based on effective duty date this record was the former commander), civilian director (3), commander of another/deployed unit (4) and commander at the NAF/Center or MAJCOM level (5). I also created a code to indicate a unit with no commander in the survey population. I used these codes to create a smaller commander/director-only dataset containing records with codes 1 and 3 (n = 2,102) for analysis. Within this dataset 110 (5.2%) respondents started but did not finish the survey and 973 (46.3%) respondents finished the survey, for a total response rate of 51.5%. Finally, the study population, a subset of the commander/director-only population, included only those cases with complete data across all variables of interest in the research model. The study population of 256 cases thus had 100% participation in the survey and both the supervisor and subordinates assessed the commander's unit on MTS effectiveness and the commander on leader rhetoric and intergroup relational identity, respectively.

Tables 1 and 2 depict survey, commander/director-only and study population demographics. The commander/director-only and study populations included slightly more white males and were more senior in rank than the full survey population with the latter being in line with demographics of commanders at the squadron-level and above. I conducted F-tests to check the assumption of equal variances within each population for demographic categories of gender, race, Hispanic, and rank. I then conducted T-tests assuming equal or unequal variance, as appropriate, based on the results of the F-test. In all cases, the two-tail p ($T \leq t$) was greater than 0.05 indicating we fail to reject H_0 and the differences in means are not significant between those who responded to the survey and those who did not within each population.

Measures

To avoid common method bias and single instrument/single informant bias, I collected archival demographic data from multiple sources and multiple actors responded to different elements of the survey. Obtaining predictor and criterion variables from different sources is one method recommended (Podsakoff et al., 2003; Podsakoff et al., 2011) to control for potential method bias in research designs. As noted previously, individual and unit demographic data from MilPDS, DCPDS, and MPES served as the basis for identifying individuals and units in the survey population. The Functional Diversity measure used data from Unit Manning Documents (UMDs) to calculate a measure of diversity using the number and type of AFSCs assigned within a unit. The Air Force Civil Engineer Center provided real property data for Air Force owned and operated installations. I computed reliability measures (Cronbach α) for each survey

Table 1. Population Demographics – Gender and Ethnicity

Variable		Survey Population				Commanders/Directors-Only Pop.				Study Pop.	
		Received		Responded		Received		Responded		Responded	
		N	%	N	%	N	%	N	%	N	%
Gender	Male	14432	77.1	1331	80.1	1711	81.4	892	82.4	223	87.1
	Female	4290	22.9	4905	19.9	391	18.6	191	17.6	33	12.9
	Total	18722	100.0	6236	100.0	2102	100.0	1083	100.0	256	100.0
Ethnic	White	14501	77.5	5007	80.3	1744	83.0	910	84.0	213	83.2
	American Indian/Native Alaskan	110	0.6	28	0.5	7	0.3	2	0.2	0	0
	Asian	680	3.6	175	2.8	64	3.0	29	2.7	7	2.7
	Black or African American	1615	8.6	443	7.1	125	5.9	51	4.7	13	5.1
	Declined	925	4.9	311	5.0	116	5.5	68	6.3	18	7
	Hispanic/Latino	268	1.4	70	1.1	0	0.0	0	0.0	0	0
	Identified more than one race	490	2.6	169	2.7	36	1.7	18	1.7	4	1.6
	Native Hawaiian/Other Pac Islander	133	0.7	33	0.5	10	0.5	5	0.5	1	0.4
	Total	18722	100.0	6236	100.0	2102	99.9	1083	100.0	256	100.0

Table 2. Population Demographics – Hispanic and Rank

Variable		Survey Population				Commanders/Directors-Only Pop.				Study Pop.	
		Received		Responded		Received		Responded		Responded	
		N	%	N	%	N	%	N	%	N	%
<i>Hispanic</i>	Hispanic or Latino	1078	5.8	484	7.8	91	4.3	46	4.2	11	11.3
	Decline to respond	1537	8.2	330	5.3	245	11.7	129	11.9	29	4.3
	Not Hispanic or Latino	16107	86.0	5422	86.9	1766	84.0	908	83.9	216	84.4
	Total	18722	100.0	6236	100.0	2102	100.0	1083	100.0	256	100.0
<i>Rank</i>	Airman	11	0.1	1	0.0	N/A	N/A	N/A	N/A	N/A	N/A
	Junior Noncommissioned Officer	240	1.2	42	0.7	N/A	N/A	N/A	N/A	N/A	N/A
	Senior Noncommissioned Officer	3058	16.3	1084	17.4	N/A	N/A	N/A	N/A	N/A	N/A
	Company Grade	4552	24.3	1127	18.1	2	0.1	1	0.1	0	0
	Field Grade	6040	32.3	2246	36.0	1503	71.5	784	72.4	237	92.6
	Senior / General Officer	896	4.8	393	6.3	571	27.2	276	25.5	19	7.4
	Civilian	3925	21.0	1343	21.5	26	1.2	22	2.0	0	0
	Total	18722	100.0	6236	100.0	2102	100.0	1083	100.0	256	100.0

scale to assess reliability of the construct. A Cronbach α value greater than or equal to 0.7 indicates acceptable levels of construct reliability.

MTS Effectiveness. Superior commanders assessed the effectiveness of their subordinate units using the four Major Graded Areas in Air Force Instruction 1-2, *Commander's Responsibilities*. The four areas are *Executing the Mission*, *Leading People*, *Managing Resources*, and *Improving the Unit*. Superior commanders were asked to rate the extent to which their subordinate units effectively executed the mission, led people, managed resources, and improved the unit using a 5-point Likert scale ranging from 1 = *Unsatisfactory* to 5 = *Outstanding*. This scale is understandable to Air Force commanders as it is the rating scale commonly used to evaluate units against these criteria. This section of the survey was designed to be completed by the superior commander of units assigned to them. If the individual identified to receive a survey invitation was assigned at the group level or above, their contact list information included the list of units to be rated by the commander since at this point in the process I did not have personnel data associated with email contact lists and could not distinguish which recipient of 5, for example, was the commander of the unit. The addition of screening questions should have prevented non-commanders from accessing this portion of the survey. However, in some cases, inaccurate responses to screening questions allowed a survey participant to inappropriately assess subordinate unit performance. These instances were ignored and only MTS Effectiveness ratings from records identified as the unit commander were retained and attributed to units as appropriate. Cronbach α for the study population was 0.79.

Intergroup Relational Identity. The 5-item empirically validated intergroup relational identity scale (Rast et al., 2019) was adapted for an Air Force setting by replacing [ingroup] and [outgroup] with appropriate terms. Sample questions included “The relationship my [component team] has with the [MTS] is part of who we are” and “The relationship between the [component team] and the [MTS] is important to what kind of [Superordinate Group] we are.” The component team, MTS and Superordinate Group were specific to each individual and varied by their level and unit of assignment. For example, an individual assigned to the 355th Civil Engineer Squadron (component team) received the following prompts: [component team] = squadron, [MTS] = 355th Mission Support Group and [Superordinate Group] = group. Cronbach’s α using the full survey population was 0.95. For the commander/director-only dataset, scores from subordinates were attributed to their commander and aggregated to create a new commander-specific intergroup relational identity variable. I computed $r_{wg(ij)}$ to generate a measure of the agreement within the group and ensure the result met the benchmark cut-off value of 0.7 to support aggregation to the team level (LeBreton & Senter, 2008). The $r_{wg(ij)}$ median of 0.94 exceeded this cut-off and Cronbach α for this aggregated construct was 0.94 in the study population.

Leader Rhetoric. The 9-item leader rhetoric scale (Rast et al., 2019) was also adapted to fit an Air Force setting. For example, the item “This leader stresses that native Dutch citizens and non-Dutch immigrants work together while preserving their distinct and separate ethnic identities” was changed to read “My commander stresses that members of our unit work with those outside our unit while preserving each unit's distinct and separate identities.” Both intergroup relational identity and leader rhetoric scales

were answered by subordinates and capture their assessment of their immediate superior commander using 7-point Likert scales ranging from 1 = Strongly disagree to 7 = Strongly agree. Cronbach α for the full dataset was 0.93. Subordinate scores were attributed to their commander and aggregated to create a new commander-specific leader rhetoric variable with $r_{wg(ij)}$ median of 0.96 and Cronbach α of 0.92 in the study population.

Boundary Spanning. Boundary spanning was measured as a self-assessment using the 4-item empirically validated scale developed by Faraj and Yan (2009) adapted from a team focus to an individual assessment and modified for an Air Force setting. For example, the item “To what extent does the team encourage its members to solicit information and resources from elsewhere in and/or beyond the division?” was modified to read “To what extent do I solicit information and resources from elsewhere in and/or beyond my unit?” All items were assessed using a 7-point Likert scale ranging from 1 = *Never* to 7 = *Every Time*. Cronbach α was 0.82 in the study population.

Systems Thinking. Systems thinking was measured as a self-assessment using the 20-item scale developed by Moore, Dolansky, Singh, Palmieri, & Alemi (2010). Survey respondents were asked to indicate their frequency of agreement with each statement using a 7-point Likert scale ranging from 1 = *Never* to 7 = *Every Time*. Sample items include “I seek everyone’s view of the situation,” “I think understanding how the chain of events occur is critical,” and “I keep in mind that proposed changes can affect the whole system.” The scoring scale was modified from the 0 to 4 scale used by Moore and colleagues (2010) to 1 to 7 to be consistent with the scale employed used for the other survey measures. Cronbach α for the study population was 0.92.

Leader Experience. Diversity of experience variables were computed using personnel demographic data from MilPDS and DCPDS. Separate variables for the number of commands of assignment, the level of commands of assignment, the number of assignments within an individual's functional specialty (AFSC), the number of joint assignments, and the number of days deployed were developed for each individual invited to participate in the survey. These items were not available for civilian personnel and as such, 26 respondents in the commander/director-only data set are missing data for these variables.

Table 3. Descriptive Statistics for Leader Experience Variables in Study Population

Variable	Minimum	Maximum	Mean	Standard Deviation
Number of commands	3	9	5.41	1.38
Highest level of assignments*	3	8	7.23	1.53
Number of core career field assignments	0	20	11.02	4.38
Number of joint assignments	1	7	2.00	1.22
Number of days deployed (days)	0	363	26.14	64.22

Note: * = categorical variable

Table 3 above depicts the descriptive statistics for leader experience variables in the study population. The level of commands of assignment variable was operationalized as a categorical variable ranging from 1 = not applicable or no longer used to 8 = service on a joint staff or the Headquarters Air Force staff.

Functional Diversity. Each Air Force unit has a UMD listing total personnel authorized by AFSC. I used UMDs for each unit to compute an index of functional diversity. In line with recommendations from Harrison and Klein (2007) I computed a modified Blau (1977) index, $Blau_N$, taking into account group size (Biemann and Kearney, 2010). The index used the total number of authorizations in each unit as well as the number of each AFSC authorized in the unit (e.g., 32EX for civil engineer officers, 3E7XX for enlisted fire fighters) to create a measure of the degree of homogeneity or heterogeneity (variety) in each unit. The index ranges from zero (no diversity among AFSCs) to one (maximum diversity of AFSCs). The diversity index for groups and wings included the group staff and wing staff authorizations, as appropriate, plus all subordinate units (component teams) in order to capture the diversity of the component teams within the MTS led by those commanders. As a final note, the number and type of personnel authorized on a UMD is often not equal to the number and type of personnel assigned to a unit due to manning shortages, personnel turnover, and availability. The minimum value was 0.04, maximum 1.0, mean 0.65, and standard deviation 0.23.

Leader and Team Identification. Leader identification was measured as a self-assessment examining an individual's identification with their component team (unit) and their MTS using the 6 items in Mael and Ashforth's (1992) empirically validated organizational identification scale. Each item was assessed using a 7-item Likert scale ranging from 1 = *Strongly Disagree* to 7 = *Strongly Agree*. The scoring scale was reversed from the original (i.e., 1 = *Strongly Agree*) to align with the rest of the survey measures. Sample items include "When someone criticizes my [unit] it feels like a personal insult" and "My [unit's] successes are my successes" where [unit] is replaced by

squadron in squadron identification, group in group identification, and so forth.

Cronbach's α computed on the full survey population for each level of identification, e.g., flight (0.81), squadron (0.71), group (0.85), wing (0.84), NAF/Center (0.88) and MAJCOM/DRU (0.97) were within the acceptable range. I built new variables to capture leader component team identification and leader MTS identification using a respondent's survey responses but agnostic as to the foci of identification. In other words, the variable representing leader component team identification might contain scores from responses to the squadron identification scale for a squadron commander while it would contain responses with a focus on group identification scale for a group commander. The variable representing leader MTS identification was constructed in a similar fashion. Cronbach's α for these two constructs were 0.74 and 0.85, respectively.

The team identification variables reflect subordinate views of identification with their component team and MTS. These variables excluded scores from the unit commander as the leader's identification is captured above in the leader identification variables. I aggregated at the component team and MTS level and then attributed as appropriate to the unit commander. Cronbach α for team component identification was 0.78 and for team MTS identification was 0.92 and the $r_{wg(ij)}$ median was 0.93 and 0.85, respectively.

Control Variables. Control variables included data elements contained in archival data sets. First, collocation with a parent group or wing may influence a commander's view of the importance of boundary spanning activities. Unit collocation was operationalized as a dichotomous variable, with 0 indicating no collocation and 1

representing collocation. As shown in Table 4 below, most units in the study population are collocated with their parent unit.

Table 4. Descriptive Statistics for Control Variables in Study Population

Variable	Minimum	Maximum	Mean	Standard Deviation
Parent unit collocation*	0	1	0.86	0.34
Gender*	0	1	0.13	0.34
Tenure (Days)	10	1,102	439.21	245.32
Number of acres on an installation (Acres)	916	3,124,905	98,179.3	343,799.30
Number of buildings and structures	342	5,765	2,050.5	1,578.80
Total square footage of buildings and structures (Square Feet)	1,215,743	43,595,961	10,707,483.0	11,077,620.30

Note: * = categorical variable

Second, as most of the respondents are likely to be male, gender was a control variable to avoid masking significant results from female respondents. Gender was operationalized with 0 representing males and 1 for females and as shown in Table 4 above, the majority of the study population was male. Third, tenure in current position assessed the influence temporal experience had on strength of identification with the component team or MTS and if a short observation period influenced ratings from subordinates or supervisors. The variable was operationalized as the number of days in the respondent's current position. Most command tours for military personnel are two years in duration indicating the average commander was in their second year of command (Table 4).

Next, host units provide installation support services to all units assigned to a given installation. Wings and subordinate support units with host responsibilities may have a different view of boundary spanning responsibilities as compared to wings without host unit responsibilities, particularly when the number of units supported by the host unit is large. This variable was operationalized as a categorical variable, with 1 indicating the wing had both mission and host unit responsibilities, 2 reflecting the wing had only mission responsibilities, and 3 representing wings whose only function was to serve as host for the installation. For the study population, there were 152 commanders who were part of a wing with both mission and host responsibilities, 84 commanders who were part of a wing with mission-only responsibilities, and 20 commanders who were part of a wing with host-only responsibilities. Finally, while most of the respondents were likely to be officers, enlisted members and civilian employees lead some flights, squadrons, and groups. Controlling for the leader's status (i.e., officer, enlisted, or civilian) will avoid masking any significant results associated with enlisted or civilian leaders. This control was dropped during the analysis phase given the commander/director-only dataset was almost exclusively officers, with just 26 civilian directors and no enlisted members in the population for analysis.

Key data elements from each Air Force installation hosting an Air Force unit were collected from real property records and intended to provide a measure of the scope, scale, and complexity of operating from a particular installation. Size (acreage), number of buildings and structures, and total square footage were collected and available as potential control variables. Descriptive statistics are reported above in Table 4.

CHAPTER 4: FINDINGS, RESULTS, AND INTERPRETATIONS

Data Analysis Preparation

I used several analytic tools (Mplus version 7.4, SPSS version 27, Excel for Mac version 16.46, and R studio version 1.3.1093) to clean and tidy data both before survey administration and after I received the merged dataset. I used SPSS to run descriptives, frequencies, and perform reliability calculations on study variables as needed. As noted in chapter 3, I checked the dataset for errors and created new variables, when required, to permit further categorization and analysis. Once the full dataset was cleaned, I computed aggregated variables to be attributed to the unit commander for leader rhetoric, intergroup relational identity, team component team identification, and team MTS identification. Finally, I created the analysis dataset of 2,102 commanders and civilian directors by selecting records with the appropriate commander-type code.

I used SPSS to conduct a principal component analysis (PCA) of all self-assessed variables in the analysis and study population dataset to ensure latent variables built from survey responses loaded appropriately against their associated construct. The four variables loaded appropriately against four factors with boundary spanning, leader component team identification, and leader MTS identification loading with all factor values > 0.5 . One systems thinking item, question 12, had a factor value of 0.486. I removed this item and ran the PCA a second time. On this second run question 9 was below the cutoff at 0.498. I removed this item, reran the PCA, and all items loaded appropriately to 4 factors with values > 0.5 , leaving a systems thinking scale with 18 items.

While reviewing the data I noted incomplete responses in some scale variables. The systems thinking scale had the most incomplete answers, with the majority of incomplete responses answering only ten of the 20 questions. For survey purposes I split the systems thinking scale into two screens in the Qualtrics survey platform, each containing ten questions, to enhance ease of use for respondents. It is possible some respondents skipped ahead after answering the first ten questions to see what remained in the survey and did not return to answer the remaining systems thinking questions. I applied a two-thirds of recorded responses cut-off to all survey variables and ignored those not meeting that standard, calculated new scale means and standard deviations, and recomputed aggregate variables.

Given the research model examines leadership within a hierarchical structure, an individual leader is acting at times as the commander of his or her unit and at others as a subordinate commander within a group or wing. Membership in the MTS likely influences not only their actions but their subordinates as well. Therefore, nonindependence among the respondents in the study population cannot be assumed. Multilevel models appropriately account for the hierarchical structure and nested relationships inherent in the study population, use clustering to avoid potential standard error bias, and enable researchers to examine interactions across levels if desired (Geiser, 2013). I used Mplus version 7.4 (Muthén & Muthén, 2015) to perform the multilevel analysis. MTS membership served as the clustering variable at level 2. All study variables resided at level 1, as depicted in Figure 3, and included the control variables gender and parent unit collocation. Initial analysis showed the remaining control variables had no significant effect when included in hypothesis testing for the

commander/director-only population. One additional control, below-the-zone promotion, was added due to its significant effect in several hypotheses when using the commander/director-only population.

Results and Interpretations

The correlation matrix for study variables at Table 5 depicts the mean, standard deviation, n, Pearson's correlation and level of significance. While the commander/director-only dataset contained 2,102 observations, missing values reduced the sample size for regression and multi-level models to 114 for MTS effectiveness and 256 for the remaining study variables.

I built and tested a series of six path models depicted in Tables 6, 7, and 8 with the dependent variable as MTS effectiveness, intergroup relational identity, and leader rhetoric, respectively. I first examined the relationship between MTS effectiveness as the dependent variable and study covariates (Model 1). Parent unit collocation ($\beta = -.05$, $SE = .09$, $p = \text{n.s.}$), gender ($\beta = -.02$, $SE = .08$, $p = \text{n.s.}$) and below-the-zone promotion ($\beta = .14$, $SE = .09$, $p = \text{n.s.}$) were all not significant. I then examined all linear relationships in the model (Model 2). Results showed the only significant linear relationship across all three dependent variables was the expected positive relationship between leader rhetoric and the development of an intergroup relational identity ($\beta = .50$, $SE = .06$, $p < .001$) (Hogg et al., 2012; Rast et al., 2018).

Table 5. Correlation Matrix for Study Population

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13
1. MTS Effectiveness	.79 ^b												
2. Intergroup Relational Identity	0.06	.94 ^b											
3. Leader Rhetoric	-0.02	.50**	.92 ^b										
4. Systems Thinking	-0.08	-0.03	-0.02	.92 ^b									
5. Boundary Spanning	-0.12	-0.10	-0.08	.39**	.82 ^b								
6. Leader Component Team Identification	0.02	-0.03	-0.02	.13*	0.05	.74 ^b							
7. Leader MTS Identification	-0.04	0.05	-0.02	.16*	0.01	.47**	.85 ^b						
8. Team Component Team Identification	0.03	.17**	.41**	0.04	-0.04	0.01	-0.09	.78 ^b					
9. Team MTS Identification	0.06	.35**	.26**	-0.01	-0.06	0.04	0.11	.30**	.92 ^b				
10. Functional Diversity	0.12	-0.05	0.00	0.03	0.08	0.08	0.10	0.02	-0.08	1.0			
11. Parent Unit Collocation ^c	-0.05	0.11	-0.02	0.11	0.11	-0.07	-0.01	-0.05	.12*	-0.12	1.0		
12. Gender ^c	-0.01	0.07	0.03	-0.01	0.09	0.09	-0.05	-0.05	0.07	0.02	0.05	1.0	
13. Below-the- Zone ^c Promotion	0.14	0.06	0.05	0.07	0.06	-0.01	-0.02	0.11	0.11	.24**	-0.05	-0.03	1.0
Mean	4.36	5.32	5.64	5.92	5.35	6.04	5.12	5.82	4.74	0.65	0.86	0.13	0.21
SD	0.57	1.03	0.83	0.57	0.90	0.74	0.99	0.83	0.77	0.23	0.34	0.34	0.41

Notes:

^an=256 individuals for variables 2 - 13; n=114 individuals for variable 1; Sig. (two-tailed) * $p \leq 0.05$, ** $p \leq 0.01$

^bCronbach Alpha

^cCategorical Variables

Table 6. Direct and Indirect Effects of Study Variables on MTS Effectiveness

<i>Predictor</i>	<i>Model 1</i>		<i>Model 2</i>		<i>Model 3</i>		<i>Model 4</i>		<i>Model 5</i>		<i>Model 6</i>	
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant	.07	.25	.15	.25	.15	.25	.15	.26	.15	.26	.15	.26
<i>Covariates</i>												
Parent Unit Collocation	-.05	.09	-.06	.09	-.06	.09	-.06	.09	-.06	.09	-.06	.09
Gender	-.02	.08	-.04	.09	-.04	.09	-.04	.09	-.05	.09	-.05	.09
Below the Zone Promotion	.14	.09	.10	.08	.10	.08	.10	.08	.09	.08	.09	.08
<i>Linear</i>												
Intergroup Relational Identity	--	--	.13	.12	.13	.12	.13	.12	.09	.13	.09	.13
Functional Diversity	--	--	.12	.10	.12	.10	.12	.10	.12	.10	.12	.10
Team CT Identification	--	--	.06	.09	.06	.09	.06	.09	.03	.11	.03	.11
Team MTS Identification	--	--	.04	.13	.04	.13	.04	.13	.06	.12	.06	.10
Leader Rhetoric	--	--	-.17	.15	-.17	.15	-.17	.15	-.17	.15	-.17	.15
Boundary Spanning	--	--	--	--	--	--	--	--	--	--	--	--
Leader CT Identification	--	--	.09	.10	.09	.10	.09	.10	.10	.10	.10	.10
Leader MTS Identification	--	--	-.06	.11	-.06	.11	-.06	.11	-.07	.11	-.07	.11
Systems Thinking	--	--	-.11	.10	-.11	.10	-.11	.10	-.10	.11	-.10	.11
<i>Interaction</i>												
Leader Rhetoric x Boundary Spanning	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Functional Diversity	--	--	--	--	--	--	.00	.09	--	--	-.01	.10
Intergroup Relational Identity x Team CT Identification	--	--	--	--	--	--	--	--	-.03	.08	-.03	.08
Intergroup Relational Identity x Team MTS Identification	--	--	--	--	--	--	--	--	-.13	.17	-.13	.17
R²	.02		.09		.09		.09		.11		.11	
ΔR^2	--		.07		0		0		.02		.02	

Note: * $p < .05$ ** $p < .01$ $N = 256$. Clustering variable is Member MTS. For model 1, $N = 114$ with 82 clusters. For models 2-6, $N = 256$ with 180 clusters.

Table 7. Direct and Indirect Effects of Study Variables on Intergroup Relational Identity

<i>Predictor</i>	<i>Model 1</i>		<i>Model 2</i>		<i>Model 3</i>		<i>Model 4</i>		<i>Model 5</i>		<i>Model 6</i>	
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant			-.04	.06	-.03	.06	-.04	.06	-.04	.06	-.04	.06
<i>Linear</i>												
Intergroup Relational Identity	--	--	--	--	--	--	--	--	--	--	--	--
Functional Diversity	--	--	--	--	--	--	--	--	--	--	--	--
Team CT Identification	--	--	--	--	--	--	--	--	--	--	--	--
Team MTS Identification	--	--	--	--	--	--	--	--	--	--	--	--
Leader Rhetoric	--	--	.50***	.06	.49***	.06	.50***	.06	.50***	.06	.50***	.06
Boundary Spanning	--	--	-.06	.06	-.06	.06	-.06	.06	-.06	.06	-.06	.06
Leader CT Identification	--	--	-.06	.07	-.06	.07	-.06	.07	-.06	.07	-.06	.07
Leader MTS Identification	--	--	.09	.06	.08	.06	.09	.06	.09	.06	.08	.06
Systems Thinking	--	--	-.00	.05	-.00	.05	-.00	.05	-.00	.05	-.00	.05
<i>Interaction</i>												
Leader Rhetoric x Boundary Spanning	--	--	--	--	.04	.06	--	--	--	--	.04	.06
Intergroup Relational Identity x Functional Diversity	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Team CT Identification	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Team MTS Identification	--	--	--	--	--	--	--	--	--	--	--	--
R²	--		.26***		.25***		.26***		.26***		.25***	
ΔR^2	--		--		-.01		0		0		-.01	

Note: * $p < .05$ ** $p < .01$ *** $p < .001$ $N = 256$. Clustering variable ($n = 180$) is Member MTS.

Table 8. Direct and Indirect Effects of Study Variables on Leader Rhetoric

<i>Predictor</i>	<i>Model 1</i>		<i>Model 2</i>		<i>Model 3</i>		<i>Model 4</i>		<i>Model 5</i>		<i>Model 6</i>	
	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Constant	--	--	-.03	.06	-.03	.06	-.03	.06	-.03	.06	-.03	.06
Linear	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity	--	--	--	--	--	--	--	--	--	--	--	--
Functional Diversity	--	--	--	--	--	--	--	--	--	--	--	--
Team CT Identification	--	--	--	--	--	--	--	--	--	--	--	--
Team MTS Identification	--	--	--	--	--	--	--	--	--	--	--	--
Leader Rhetoric	--	--	--	--	--	--	--	--	--	--	--	--
Leader CT Identification	--	--	-.01	.07	-.01	.07	-.01	.07	-.01	.07	-.01	.07
Leader MTS Identification	--	--	-.02	.08	-.02	.08	-.02	.08	-.02	.08	-.02	.08
Systems Thinking	--	--	-.01	.07	-.01	.07	-.01	.07	-.01	.07	-.01	.07
Interaction	--	--	--	--	--	--	--	--	--	--	--	--
Leader Rhetoric x Boundary Spanning	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Functional Diversity	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Team CT Identification	--	--	--	--	--	--	--	--	--	--	--	--
Intergroup Relational Identity x Team MTS Identification	--	--	--	--	--	--	--	--	--	--	--	--
R²	--		.00		.00		.00		.00		.00	
ΔR^2	--		0		0		0		0		0	

Note: * $p < .05$ ** $p < .01$ $N = 256$. Clustering variable ($n = 180$) is Member MTS.

Hypothesis 1 predicted a positive indirect effect between leader rhetoric and MTS effectiveness through the development of an intergroup relational identity. I used the multistep Baron and Kenny (1986) framework for mediation to evaluate the hypothesized relationships. As shown in Model 2 (Table 6), I calculated the unmediated effect from leader rhetoric to MTS effectiveness ($\beta = -.17$, $SE = .15$, $p = n.s.$) and as shown in Model 2 (Table 7) the path from leader rhetoric to the mediator, intergroup relational identity ($\beta = .50$, $SE = .06$, $p < .001$). I also tested the path from intergroup relational identity acting as a mediator to MTS effectiveness as the outcome variable as shown in Model 2 (Table 6) ($\beta = .13$, $SE = .12$, $p = n.s.$). Results showed that while leader rhetoric significantly predicted intergroup relational identity, intergroup relational identity and leader rhetoric were not significantly related to MTS effectiveness. For a mediation effect to occur, the addition of the mediator should significantly reduce the strength of the relationship between the independent variable and the outcome variable or make the relationship insignificant. That did not occur as the addition of intergroup relational identity as a mediating variable did not change the nature of the relationship between leader rhetoric and MTS effectiveness. As a confirmation step, I used the model indirect command in Mplus to calculate the indirect effect and found it was not significant ($\beta = .07$, $SE = .06$, $p = n.s.$) as depicted in Table 9 below. The hypothesized mediation effect predicted in Hypothesis 1 therefore is not supported.

Table 9. Direct and Indirect Effects of Leader Rhetoric on Intergroup Relational Identity through Boundary Spanning

	<i>Effect</i>	<i>SE</i>
Total	-.11	.13
Total Indirect	.07	.06
Specific Indirect	.07	.06
Direct	-.17	.15

*Note: * $p < .05$ ** $p < .01$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.*

To investigate whether boundary spanning acts as moderator of the positive relationship between leader rhetoric and intergroup relational identity as predicted in Hypothesis 2, I added an interaction effect between leader rhetoric and boundary spanning (Model 3). As noted in Table 7 and discussed above, while leader rhetoric significantly predicted a positive impact on leader rhetoric, boundary spanning had a non-significant negative relationship with intergroup relational identity ($\beta = -.06$, $SE = .06$, $p = n.s$), the interaction was not significant ($\beta = .04$, $SE = .06$, $p = n.s$), and R^2 decreased by .01.

I did, however, find significant conditional effects as depicted in Table 10 and illustrated in Figure 4. Using a simple slope calculation via the model constraint command in Mplus I calculated values at one standard deviation below (low) and one standard deviation above (high) the boundary spanning mean. Results show that as levels of boundary spanning increase the effect of leader rhetoric on intergroup relational identity increases. The slope for low levels of boundary spanning is higher than the slope for high levels of boundary spanning. While this pattern is less inconsistent with the

predicted relationship in Hypothesis 2, the results do not support the theorized relationship in this hypothesis.

Table 10. Conditional Effects of Leader Rhetoric on Intergroup Relational Identity

	Effect	SE	p
Low Boundary Spanning (-1 SD)	.39	.12	.001
Mean Boundary Spanning	.44	.06	.000
High Boundary Spanning (+1 SD)	.49	.07	.000

Notes: N = 256. Values are standardized coefficients.

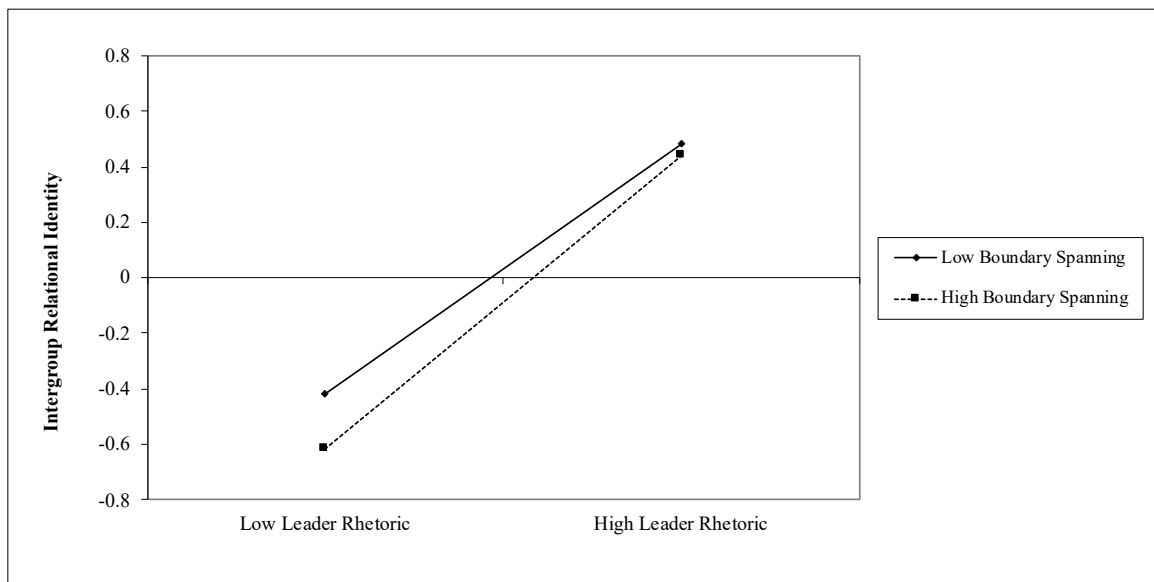


Figure 4. Moderating Effect of Boundary Spanning on the Relationship Between Leader Rhetoric and Intergroup Relational Identity

A series of leader experience characteristics were predicted to have an indirect effect on the development of an intergroup relational identity through leader rhetoric as moderated by boundary spanning leadership such that leaders with more diverse career paths, higher systems thinking skills, and stronger identification with the MTS would

have a more positive impact on the development of this relationship. While Model 3 results showed there were several leader experience variables with significant direct relationships to MTS effectiveness (highest command level at the installation, DRU, and Field Operating Agency plus number of units and days deployed), intergroup relational identity (highest command level at a DRU and number of joint tours) and leader rhetoric (number of assignments in core career field) they were dropped from the analysis after further investigation of the study population contributing to those findings. I built crosstabs in SPSS to examine the number of cases contributing to the significant outcomes since the results were surprising based on my experience. The dummy variable capturing the highest level of assignment residing at a DRU represented just one case and therefore the highly significant relationship with MTS effectiveness ($\beta = -.11$, $SE = .02$, $p < .001$) and intergroup relational identity ($\beta = .05$, $SE = .01$, $p < .001$) was not representative of the study population. Hypothesis 3a therefore was dropped from further consideration and evaluation.

While systems thinking had a highly significant positive correlation with boundary spanning ($r = .39$, $p < .01$), it did not have the hypothesized indirect effect on intergroup relational identity. As shown in Tables 7 and 8 (Model 3), systems thinking had an insignificant negative relationship with both intergroup relational identity ($\beta = -.00$, $SE = .05$, $p = \text{n.s.}$) and leader rhetoric ($\beta = -.01$, $SE = .07$, $p = \text{n.s.}$). As expected based on these results, when testing for indirect effects using the model indirect command in Mplus, there were no significant direct or indirect effects as reported below in Table 11 and thus Hypothesis 3b is not supported.

Table 11. Direct and Indirect Effects of Systems Thinking on Leader Rhetoric and Intergroup Relational Identity through Boundary Spanning

	<i>Effect</i>	<i>SE</i>
Total	-.01	.06
Total Indirect	.01	.01
Specific Indirect	.01	.03
Direct	-.00	.05

*Note: * $p < .05$ ** $p < .01$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.*

Finally, Hypothesis 3c predicted leaders who more strongly identify with the MTS as opposed to their component team would have a positive indirect effect on the development of an intergroup relational identity through leader rhetoric as moderated by boundary spanning leadership. As shown in Tables 7 and 8 (Model 3), leader component team identification was not significantly related to either intergroup relational identity ($\beta = -.06$, $SE = .07$, $p = \text{n.s.}$) or leader rhetoric ($\beta = -.01$, $SE = .07$, $p = \text{n.s.}$) and no significant indirect effects were found as reported in Table 12. Leader MTS identification was similarly not significantly related to intergroup relational identity ($\beta = .08$, $SE = .06$, $p = \text{n.s.}$) or leader rhetoric ($\beta = -.02$, $SE = .08$, $p = \text{n.s.}$) and as shown in Table 13, the indirect effect was nonsignificant. While leader MTS identification was positively related and leader component team identification negatively related to intergroup relational identity, the nonsignificant finding prevents drawing any conclusions in support of this hypothesis and therefore the results do not support Hypothesis 3c.

Table 12. Direct and Indirect Effects of Leader Component Team Identification on Leader Rhetoric and Intergroup Relational Identity through Boundary Spanning

	<i>Effect</i>	<i>SE</i>
Total	-.06	.08
Total Indirect	-.00	.04
Specific Indirect	-.00	.04
Direct	-.06	.07

*Note: * $p < .05$ ** $p < .01$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.*

Table 13. Direct and Indirect Effects of Leader MTS Identification on Leader Rhetoric and Intergroup Relational Identity through Boundary Spanning

	<i>Effect</i>	<i>SE</i>
Total	.07	.07
Total Indirect	-.01	.04
Specific Indirect	-.01	.04
Direct	.08	.06

*Note: * $p < .05$ ** $p < .01$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.*

Hypothesis 4 examined component team and MTS identification from a team perspective. To investigate whether identification at the MTS level has a stronger impact on the indirect effect of leader rhetoric defining group identity as an intergroup collaborative relationship leading to higher levels of MTS effectiveness, I tested a series of path models (Model 5). Tables 6, 7, and 8 show the result of simultaneously testing leader component team identification and leader MTS identification as antecedents to leader rhetoric plus team component team identification and team MTS identification acting as moderators for the relationship between intergroup relational identity and MTS effectiveness. I also tested four additional cases as shown in Tables 14 and 15, first assessing leader component team identification alone paired with either team component

team identification or team MTS identification. I then examined the same with leader MTS identification matched with team component team identification and team MTS identification. The results from this 2x2 analysis, reported in Tables 14 and 15, show all relationships were nonsignificant. As expected based on these results, there were no significant relationships identified in the full model with all four variables included (Model 5) and therefore Hypothesis 4 is not supported.

Table 14. Direct and Indirect Effect of Leader Component Team Identification and Team Identification on Intergroup Relational Identity and MTS Effectiveness

<i>Predictor</i>	Leader Component Team Identification and Team Component Team Identification				Leader Component Team Identification and Team MTS Identification			
	Intergroup Relational Identity		MTS Effectiveness		Intergroup Relational Identity		MTS Effectiveness	
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant	-.04	.06	.17	.26	-.04	.06	.16	.25
Intergroup Relational Identity			.14	.12			.08	.13
Leader Component Team Identification	-.02	.06	.08	.09	-.02	.06	.08	.09
Team Component Team Identification			.05	.11				
Team MTS Identification							.06	.12
Intergroup Relational Identity x Team Component Team Identification			-.05	.08				
Intergroup Relational Identity x Team MTS Identification							-.15	.16
R²	.25***		.09		.25***		.08	

Note: * $p < .05$ ** $p < .01$ *** $p < .001$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.

Table 15. Direct and Indirect Effect of Leader MTS Identification and Team Identification on Intergroup Relational Identity and MTS Effectiveness

<i>Predictor</i>	Leader MTS Identification and Team Component Team Identification				Leader MTS Identification and Team MTS Identification			
	Intergroup Relational Identity		MTS Effectiveness		Intergroup Relational Identity		MTS Effectiveness	
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant	-.05	.06	.19	.26	-.05	.06	.20	.25
Intergroup Relational Identity			.13	.12			.07	.13
Leader MTS Identification	.06	.05	-.08	.10	.06	.05	-.03	.10
Team Component Team Identification			.06	.10				
Team MTS Identification							.07	.12
Intergroup Relational Identity x Team Component Team Identification			-.05	.08				
Intergroup Relational Identity x Team MTS Identification							-.14	.16
R²	.25***		.10		.25***		.08	

Note: * $p < .05$ ** $p < .01$ *** $p < .001$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.

Functional diversity, represented by the variety of authorizations present in a squadron, group, or wing, was predicted to play a moderating role on the use of leader rhetoric to build an intergroup relational identity and positively impact MTS effectiveness, making the relationship stronger with higher levels of functional diversity. In other words, units with high levels of diversity present a more difficult challenge with respect to the communication and leadership needed to build and sustain an intergroup relational identity and positively influence MTS effectiveness. To test this hypothesis, I added a moderation effect between intergroup relational identity and functional diversity

in Model 4. As shown in Table 6 the moderation effect on MTS effectiveness was not significant ($\beta = .00$, $SE = .09$, $p = n.s.$). Thus, Hypothesis 5 is not supported.

Supplemental Analysis

To further investigate potentially significant influences on the demonstrated positive relationship between leader rhetoric and intergroup relational identity I built and analyzed a new model in Mplus. Both leader component team identification and leader MTS identification were significantly correlated with systems thinking ($r = .13$, $p \leq .05$ and $r = .16$, $p \leq .05$, respectively) as indicated in Table 5. In addition, boundary spanning was significantly correlated with systems thinking ($r = .39$, $p \leq .01$) and previous research (Porck & van Knippenberg, 2020; Richter et al., 2006) had shown boundary spanning should influence on MTS effectiveness. The model that follows examines potential moderation effects of leader identification and systems thinking on the demonstrated positive relationship between leader rhetoric and intergroup relational identity and the collective impact of these relationships on MTS effectiveness. It also includes boundary spanning acting as a moderator of the relationship between intergroup relational identity and MTS effectiveness. As predicted in the original research model, higher levels of leader MTS identification and systems thinking will enhance the positive relationship between leader rhetoric and intergroup relational identity and ultimately lead to higher levels of performance while stronger identification with the leader's component team will detract from these relationships and results. Boundary spanning should likewise exhibit a positive moderation effect on the relationship between intergroup relational identity and MTS effectiveness, with higher levels of boundary spanning behavior enhancing the

impact of the intergroup relational identity on MTS effectiveness. The full supplemental analysis model is illustrated below in Figure 5.

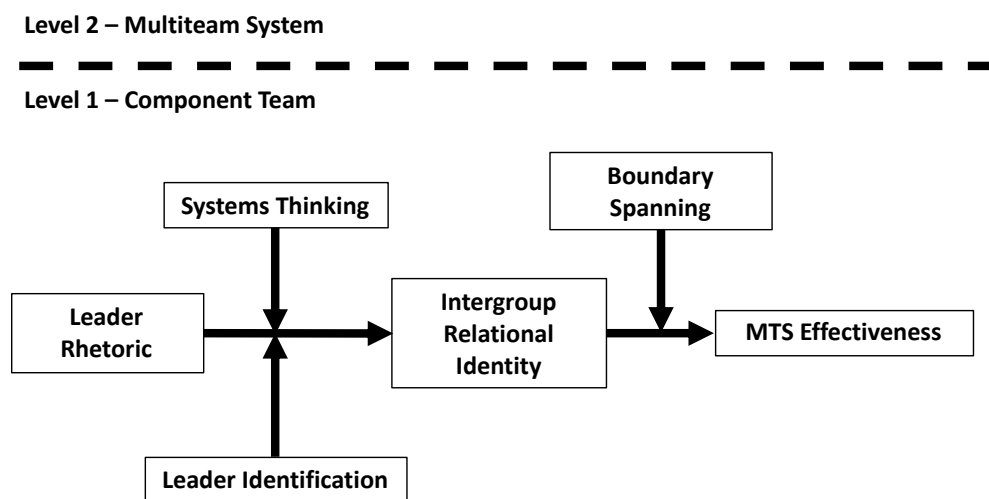


Figure 5. Supplemental Analysis Model

Systems thinking and leader identification, originally envisioned as antecedents of intergroup leadership, capturing the extent to which a leader engages in conversations and behaviors aimed at building an intergroup relational identity may instead act as moderators, influencing the quality and effectiveness of rhetoric aimed at engendering an intergroup collaborative relationship. Table 16 Model 1 depicts the results generated by this moderation model. The direct relationships between leader component team identification and intergroup relational identity ($\beta = -.07$, $SE = .09$, $p = n.s.$), leader MTS identification and intergroup relational identity ($\beta = .08$, $SE = .06$, $p = n.s.$) and systems thinking and intergroup relational identity ($\beta = -.01$, $SE = .06$, $p = n.s.$) were not significant. However, interactions between leader rhetoric and leader component team identification ($\beta = -.19$, $SE = .07$, $p < .01$) and between leader rhetoric and systems thinking ($\beta = .13$, $SE = .05$, $p = .01$) were significant as shown in Table 16 and illustrated

in Figures 6a and 6c. The interaction with leader MTS identification was just outside the cutoff for significance ($\beta = .12$, $SE = .06$, $p = .054$) as shown in Table 16 and Figure 6b.

As shown in Figure 6a, while leaders with high levels of identification with their component team may have a more positive impact on developing an intergroup relational identity with language that is perhaps understated their rhetoric is less effective when it is amplified at higher levels. On the other hand, the language used by leaders who demonstrate strong identification with their MTS is more effective toward the development of an intergroup relational identity than with leaders who have lower levels of MTS identification (Figure 6b). Finally, as shown in Figure 6c, systems thinking skills are important to the development of an intergroup relational identity in a team. A leader's ability to understand the bigger picture and translate this understanding through language that explains how various groups work together toward goal accomplishment has a more positive impact on the development of an intergroup relational identity as opposed to leaders with lower levels of system thinking skills.

Table 16. Direct and Indirect Effects of Leader Identification, Systems Thinking, and Boundary Spanning on Intergroup Relational Identity and MTS Effectiveness

<i>Predictor</i>	Model 1				Model 2				Model 3			
	Intergroup Relational Identity		MTS Effectiveness		Intergroup Relational Identity		MTS Effectiveness		Intergroup Relational Identity		MTS Effectiveness	
	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>	β	<i>SE</i>
Constant	-.05	.06	.05	.10	-.07	.06	.12	.10	-.05	.06	.12	.10
Intergroup Relational Identity	--	--	.10	.12	--	--	-.01	.12	--	--	-.01	.12
Leader Rhetoric	.52***	.06	-.09	.13	.50***	.06	-.03	.13	.52***	.06	-.03	.13
Leader Component Team Identification	-.07	.07	--	--	--	--	--	--	-.07	.07	--	--
Leader MTS Identification	.08	.06	--	--	--	--	--	--	.08	.06	--	--
Systems Thinking	-.01	.05	--	--	--	--	--	--	-.01	.05	--	--
Boundary Spanning	--	--	--	--	--	--	-.03	.13	--	--	-.10	.08
Leader Rhetoric x Leader Component Team Identification	-.19**	.07	--	--	--	--	--	--	-.19**	.07	--	--
Leader Rhetoric x Leader MTS Identification	.12	.06	--	--	--	--	--	--	.12	.06	--	--
Leader Rhetoric x Systems Thinking	.13*	.05	--	--	--	--	--	--	.13*	.05	--	--
Intergroup Relational Identity x Boundary Spanning	--	--	--	--	--	--	.21*	.11	--	--	.21*	.11
R²	.28***		.01		.25***		.06		.28***		.06	

Note: * $p < .05$ ** $p < .01$ *** $p < .001$ $N = 256$. Clustering variable ($n = 180$) is Member MTS.

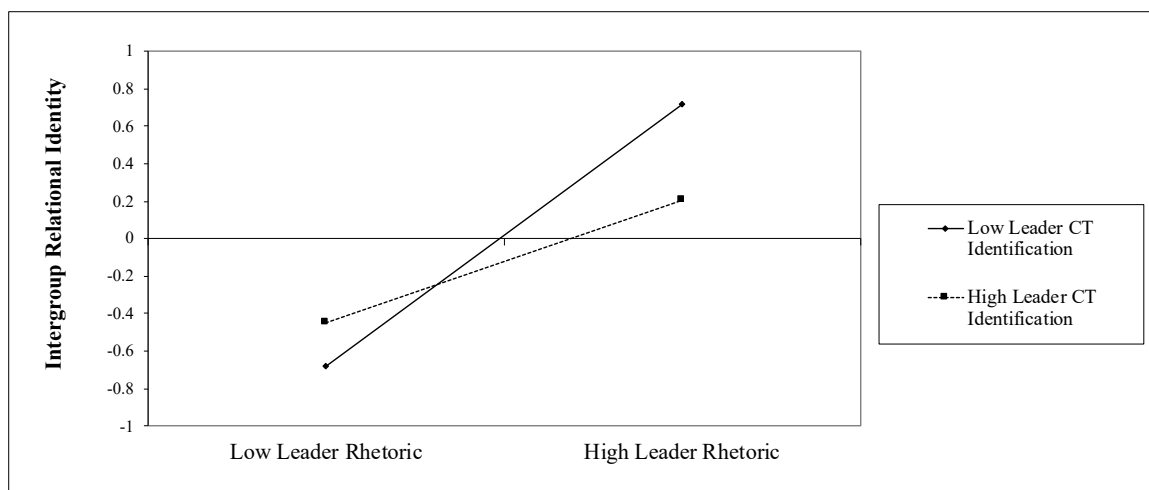


Figure 6a. Moderating Effect of Leader Component Team Identification on the Relationship between Leader Rhetoric and Intergroup Relational Identity

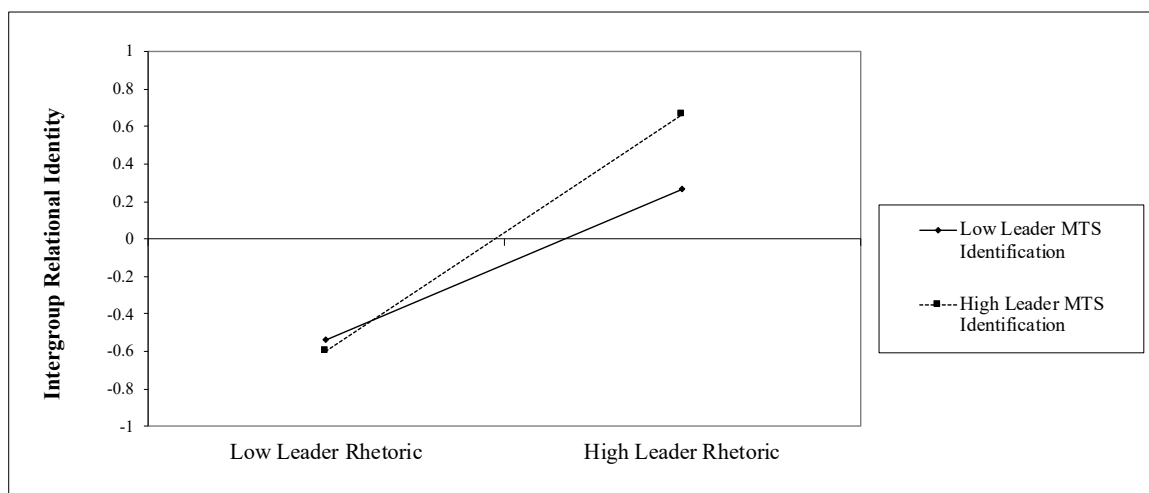


Figure 6b. Moderating Effect of Leader MTS Identification on the Relationship between Leader Rhetoric and Intergroup Relational Identity

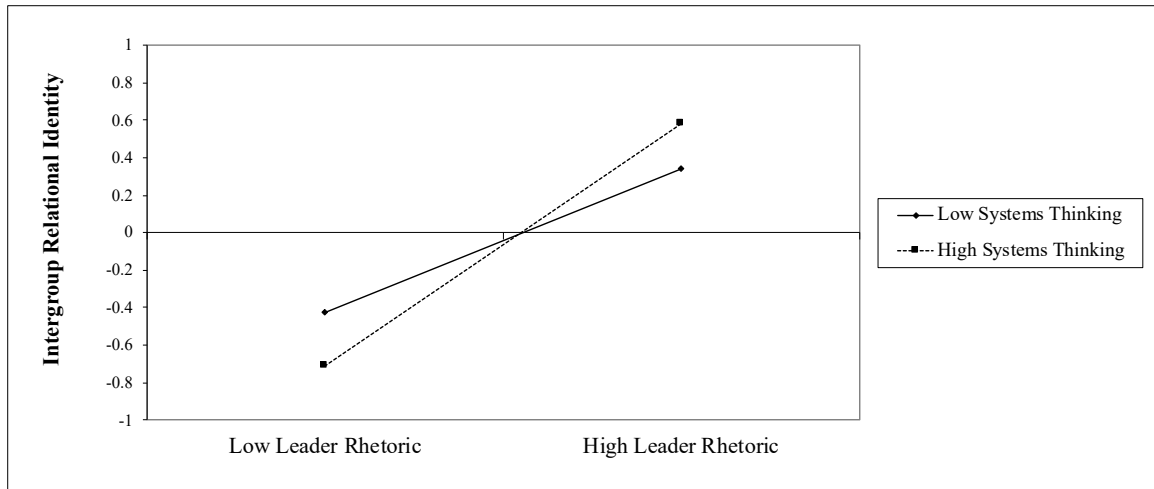


Figure 6c. Moderating Effect of Systems Thinking on the Relationship between Leader Rhetoric and Intergroup Relational Identity

As was the case in the research model findings, Table 17 shows the indirect effect of leader rhetoric through the development of an intergroup relational identity on MTS effectiveness was not significant even after the addition of the significant moderation effects outlined above and provided little additional explanatory power, with $R^2 = .01$ (Table 16, Model 1).

Table 17. Direct and Indirect Effects of Leader Rhetoric on MTS Effectiveness in Supplemental Model

	<i>Model 1</i>		<i>Model 2</i>		<i>Model 3</i>	
	<i>Effect</i>	<i>SE</i>	<i>Effect</i>	<i>SE</i>	<i>Effect</i>	<i>SE</i>
Total	-.04	.11	-.04	.11	-.04	.11
Total Indirect	.05	.06	-.00	.06	-.00	.06
Specific Indirect	.05	.06	-.00	.06	-.00	.06
Direct	-.09	.13	-.03	.13	-.03	.13

Note: * $p < .05$ ** $p < .01$ $N = 256$. Values are standardized coefficients. Clustering variable ($n = 180$) is Member MTS.

Model 2 (Table 16) reports the results of boundary spanning acting as a moderator of the relationship between intergroup relational identity and MTS effectiveness. The moderation effect was significant ($\beta = .21$, $SE = .11$, $p < .05$) and as depicted below in Figure 7, low levels of boundary spanning coupled with higher levels of intergroup relational identity results in lower levels of MTS effectiveness while the opposite is true for high levels of boundary spanning. Leaders generating high levels of intergroup relational identity but practicing low levels of boundary spanning behavior may create dissonance in their team by failing to model behaviors their rhetoric implies are important.

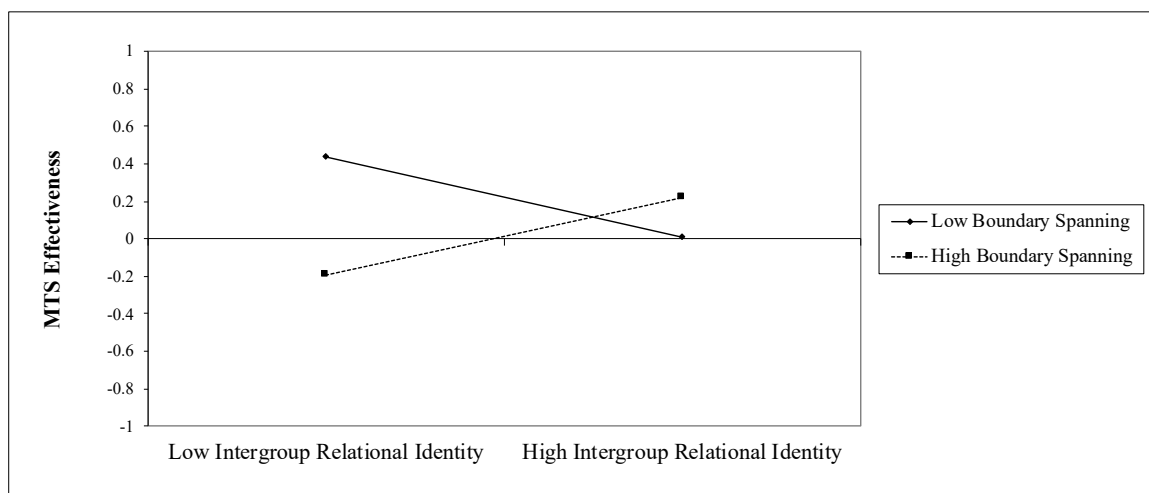


Figure 7. Moderating Effect of Boundary Spanning on the Relationship Between Intergroup Relational Identity and MTS Effectiveness

I also calculated the conditional indirect and total effects for all moderation relationships depicted in Table 16, Models 1 and 2. I used Mplus code developed to support Hayes' (2017) PROCESS model methodology (Stride, Gardner, Catley, & Thomas, 2015) to generate syntax for the 27 different conditional indirect and 27

different conditional total effects. There were no significant conditional effects across any combination of moderation variables as shown in Tables 18 to 25 (Appendix D). The results of the full supplemental model are shown in Table 16, Model 3. All significant relationships identified in Models 1 and 2 remained in Model 3.

In summary, while results did not support hypothesized relationships depicted in the research model, the findings did provide further empirical support for the positive relationship between leader rhetoric and development of an intergroup relational identity (Hogg et al., 2012; Rast et al., 2018) and provided evidence of boundary spanning's insignificant but negative conditional effect on this relationship which was contrary to expectations developed by Hogg and colleagues (2012). Supplemental analysis demonstrated the positive effect high levels of systems thinking coupled with high levels of leader rhetoric had on the development of an intergroup relational identity and demonstrated the negative impact high levels of leader component team identification can have on this same relationship. Supplemental results also supported a boundary spanning moderation effect between intergroup relational identity and MTS effectiveness, with higher levels of boundary spanning coupled with high levels of intergroup relational identity leading to higher levels of MTS effectiveness. I will address the implications of these findings for researchers and practitioners in the next chapter.

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

Introduction

This study leveraged literature from multiteam systems and intergroup leadership theory research streams to examine and further our understanding of the roles played by boundary spanning, systems thinking, identification, and functional diversity on intergroup leadership and MTS effectiveness. Results provided empirical support for the positive relationship between leader rhetoric and intergroup relational identity. While the hypothesized relationships predicted in the research model were not supported, supplemental analysis showed leaders with higher levels of systems thinking skills, when coupled with higher levels of leader rhetoric, had a more positive effect on the development of an intergroup relational identity. Leaders who had higher levels of identification with their component team when coupled with high levels of leader rhetoric were less effective in building high levels of intergroup relational identity. Results also identified a moderation effect of boundary spanning on the relationship between intergroup relational identity and MTS effectiveness, with high levels of boundary spanning coupled with high levels of intergroup relational identity leading to higher levels of MTS effectiveness.

Theoretical Implications

This study makes several important theoretical contributions. First, given the relative paucity of empirical studies exploring intergroup leadership, this study provides further evidence to support Hogg and colleagues' (2012) theory of intergroup leadership. The strength of this support is amplified by the research design, which sought the input of subordinates who work for and with the leaders they evaluated through leader rhetoric

and intergroup relational identity survey measures. This design, as opposed to capturing views of subjects participating in a simulated task or reading a hypothetical scenario with descriptions of leader rhetoric offers added depth and richness to the results.

The lack of support for boundary spanning moderating the relationship between leader rhetoric and intergroup relational identity and the finding of an insignificant but negative conditional effect was unexpected. As originally envisioned by Hogg and colleagues (2012), boundary spanning behaviors enable leaders to “walk the talk” and through their actions model behaviors necessary to establish an intergroup relational identity. Porck & van Knippenberg (2020) found middle manager boundary spanning among interdependent teams from different divisions had a positive impact on intergroup effectiveness. The research model evaluated boundary spanning tendencies of leaders but did not examine the focus of those boundary spanning activities. In other words, were the boundary spanning activities leaders envisioned when answering the survey measures beyond the component team but confined within the MTS or did they extend beyond the MTS? It is possible the way in which boundary spanning was operationalized influenced the counterintuitive finding. In addition, the cross-sectional survey design makes it difficult to assess causality as predicted in the research model. Finally, it is also possible variables not measured or included in the study influence the predicted relationship such that the moderation effect is weakened or masked. Shuffler and colleagues note that functional behaviors such as boundary spanning need additional study to clarify the true impact on MTS effectiveness (Shuffler et al., 2015).

The systems thinking scale (Moore et al., 2010) was used for the first time in an empirical study and supplemental analysis demonstrated its positive moderation effect on

the relationship between leader rhetoric and intergroup relational identity. This result should not be surprising as the ability to develop and articulate how different groups interact and work in coordination with each other to achieve team objectives would be enhanced in individuals who also possess an ability to see the big picture and understand how the larger system operates. Additional research that confirms both the significance and direction of systems thinking's influence on intergroup leadership would help determine whether this finding has broader application or is unique to this study design and organizational setting.

Finally, the supplemental analysis finding that leaders with higher levels of identification with the component team coupled with high levels of leader rhetoric had a more negative impact on the development of an intergroup relational identity is both in line with (e.g., DeChurch & Zaccaro, 2010; Connaughton et al., 2012) and contrary to (Porck et al., 2019) previous findings that examined component team and MTS identification with respect to impacts on MTS effectiveness. The study did not support a significant relationship between leader identification or team identification and MTS effectiveness, regardless of how that relationship was tested. Leader identification as an antecedent to and team identification as a moderator of intergroup leadership and its influence on MTS effectiveness were not significant. However, by envisioning leader identification as a way to reinforce and enhance the quality of the rhetoric aimed at building a collaborative intergroup relationship through development of an intergroup relational identity, I demonstrated the negative consequences of over identification with the component team. Leaders with higher levels of identification with their component team may have a more difficult time “walking the talk” and subordinates may view their

efforts to develop an intergroup relational identity as disingenuous and counterproductive.

Practical Implications

There are several practical implications for managers and leaders who are part of complex multiteam systems. First, building and enhancing systems thinking skills and providing a solid understanding of the overall system and drivers of performance will help component team leaders develop language and messaging that resonate with team members. Murase and colleagues (2014) argue more research is needed to understand the role MTS leaders have in shaping member cognition. They recommend a two-step approach, first training leaders how to make sense of the MTS environment and second, translating that training into concrete plans for component team collaboration and component team leader messaging to reinforce and deepen that knowledge and understanding throughout their team and the system as a whole. Research has demonstrated the positive effect training in MTS leadership functions has on system performance (DeChurch & Marks, 2006) and Zaccaro and colleagues (2020) note formal leadership development programs are lacking training in this area. Orientation programs that expose leaders to the full gamut of system operations can help build a foundational understanding of how each component team contributes to overall MTS goals and objectives.

Second, leader development programs for those serving in these complex MTSs should focus on how leaders can strike the right balance between engendering unit *esprit de corps* and recognizing that the unit is part of the larger MTS. Understanding the

tipping point beyond which strong identification with the component team detracts from the leader's ability to build an intergroup relational identity is especially important.

Finally, the study reinforces the importance of leaders not only developing appropriate messaging to promote an intergroup relational identity but also embodying that rhetoric through their boundary spanning activities. While the boundary spanning findings in the research model were contrary to previous research, supplemental analysis did show that an intergroup relational identity was more instrumental in producing higher levels of MTS effectiveness when boundary spanning levels were high.

Strengths and Limitations

As with any study, there are strengths and limitations in both the approach as well as the interpretation and generalizability of the results. As Burke (2014) noted, "In terms of data collection, MTSs are teams on steroids." First, to counter common method and single instrument bias, I collected data from multiple sources using archival data from multiple personnel, manpower, and real property systems to obtain demographic information about survey respondents, units, and installations. The survey data was collected from multiple actors with subordinates, supervisors, and respondents answering a series of empirically validated scales and measures. In addition, as opposed to conducting a laboratory simulation using subjects who may have no previous interaction with their study counterparts and no prior experience performing the simulated task, this study surveyed individuals who were part of teams that work together every day and thus were familiar with the individuals and units that were the focus of their assessment. This depth of understanding with leaders and teams across a variety of real-world situations

and experiences allowed respondents to evaluate leader behaviors and unit performance on a continuous basis rather than a single snapshot in time.

The study, as noted earlier, offers the first empirical examination of systems thinking and its relationship to intergroup leadership and MTS effectiveness. Given the relatively short time since the MTS was advanced as an organizing construct for teams research and intergroup leadership theory published, this is not unusual. In addition, given the challenges noted by Burke (2014) regarding field studies – data access, time, unobtrusive measurement, and interpreting deeply contextualized data – this study offers a unique addition to both research streams. It also supports Hackman’s recommendation related to bracketing in research designs by “including constructs that exist one level lower but also one level higher than those that are the main subject of study” (Hackman, 2003: 24).

The study also has its limitations. The multilevel nature of the study meant it was important not only to have a strong response rate but also a sufficient number of responses within each hierarchical relationship. The commander/director-only dataset initially contained 2,102 records but just over half of that number responded to the survey. When coupled with measures that relied on subordinate and supervisor evaluation, the study population dropped to 256 with 180 clusters (multiteam systems). The vast majority of the 256 respondents were commanders at the squadron level, with only 6 respondents each at the group and wing level having complete data across all variables used for analysis. Thus, the ability to generalize the findings to units beyond a squadron level is limited.

MTS effectiveness was measured through a supervisor assessment using a widely accepted and understood construct that evaluated unit performance against four major areas. Reliability analysis showed an acceptable result for Cronbach α ($\alpha = 0.79$) and correlations when using the commander/director-only dataset showed significant correlation between MTS effectiveness and intergroup relational identity ($r = .09$, $p \leq .05$), gender ($r = -.07$, $p \leq .05$), and below the zone promotion ($r = .15$, $p \leq .01$). The significance of these relationships did not materialize when running multilevel models using the smaller study dataset. There are two potential reasons for this. First, by limiting the study dataset to records with complete responses across all study variables I lost explanatory power emanating from partial responses. Second, it is also possible this method of operationalizing MTS effectiveness was incongruent with effectiveness measures capturing performance in a laboratory simulation. Supervisors were asked to assess performance of their subordinate units rather than those units being evaluated while performing a discrete task. As such, supervisors no doubt considered a variety of performance episodes and interactions with the subordinate unit when assessing performance. It is possible this continuous versus discrete assessment influenced the results.

As noted earlier, I spent a significant amount of time cleaning the dataset, both before and after archival data was integrated with survey data. In several instances individuals identified their level of assignment and the level of their supervisor in a way that was contrary to that reflected in the personnel system. Since the survey was constructed based on information contained in the personnel system, I identified these

instances of a mismatch and discounted them prior to aggregating team identification scores which limited the number of responses included in the aggregated variables.

Future Research

There are several potential avenues to explore in future research on both intergroup leadership and multiteam systems. First, the lack of support for boundary spanning as a moderator on the relationship between leader rhetoric and intergroup relational identity should be explored further to understand if the predicted relationship appears in other settings or if the study findings are replicated. For instance, does the relationship appear outside of an MTS setting in a single level model? If so, what is it about the MTS that mitigates the hypothesized relationship? In addition, what is driving the negative slope for low levels of boundary spanning in the moderation interaction between intergroup relational identity and boundary spanning? The bottom line is we do not yet have sufficient research in this area to fully understand the nature of these relationships and more study is required.

Second, in line with Mell and colleagues' (2020) work on identity asymmetries, the data collected as part of this study should be analyzed to determine if there is sufficient information to evaluate whether different configurations of identity foci affect MTS performance. Units with higher levels of heterogeneity (e.g., a Mission Support Group with on average 4 or more squadrons from different functional areas), based on my experience, are much more likely to contain a mixture of foci, with some squadrons adopting a stronger MTS identification while others have very strong component team identities. Different foci can lead to different drivers and reasons for collaboration (Mell et al., 2020).

In addition, in line with the meso-theory of MTS functioning developed by Luciano and colleagues (2018), countervailing forces such as high levels of goal discordancy may also highlight differences in leadership and performance. Units whose functional communities exert more influence over their policies, procedures, and norms are likely to have higher levels of goal discordancy than units without that level of “outside-the-wing” influence. For example, contracting squadrons support all units assigned to an installation by awarding services contracts and purchasing items but obtain their contracting warrants (authority to obligate the federal government) through their functional reporting chain. They are also measured both by their unit chain of command and their functional chain, creating tension between contracts awarded to large businesses familiar to customers on the installation and contracts awarded to small businesses to help their functional chain meet small business goals.

As noted above, it is important to understand the lack of significant findings associated with MTS effectiveness. Lab studies and simulations have an advantage in measuring MTS effectiveness as performance metrics are built into the simulation itself. Was the lack of significant findings due to the way in which MTS effectiveness was measured or was there another reason driving these results?

Finally, extending the level of analysis to a third level to consider the impact of location may uncover interesting dynamics influencing the external MTS environment. There are two aspects associated with location. First, installations vary in terms of size and complexity, ranging from a single mission base with few additional mission partners to complex installations supporting 100 or more units from various services and agencies. Does the complexity of the installation environment influence the strength of an

intergroup relational identity in host units? Based on my experience I would predict higher levels of intergroup relational identity are needed as the size and complexity of the installation increases. The second aspect regarding location is collocation with a parent unit. When analyzing the larger commander/director-only dataset, unit collocation was significantly correlated with several study variables. Are there differences in intergroup leadership or MTS performance for units that are not collocated with their parent unit as opposed to those that are? This is an important consideration moving forward in light of the significant spike in remote work driven by the COVID-19 pandemic. It is likely the context of remote/virtual work will have a growing impact on intergroup leadership and MTS effectiveness in the future.

Conclusion

This study leveraged MTS and intergroup leadership theory literature streams to examine the effect of identification, systems thinking, and functional diversity on intergroup leadership and multiteam system effectiveness. Although the study did not support the predicted relationships, it did empirically support the positive relationship between leader rhetoric and intergroup relational identity posited by Hogg and colleagues (2012). Supplemental analysis uncovered a new positive moderation effect of systems thinking, and a negative moderation effect generated by a leader's component team identification on the relationship between leader rhetoric and intergroup relational identity. In addition, boundary spanning acted as a positive moderator of the relationship between intergroup relational identity and MTS effectiveness when boundary spanning was high but not low. These surprising and contrary findings reinforce the need to expand research in these areas and conduct additional empirical studies that can either

replicate or refute this study's findings. The study breaks new ground by introducing systems thinking as a variable of interest when evaluating intergroup leadership in an MTS and, as one of the larger field studies conducted in this area, highlights the strengths and challenges associated with data collection and multilevel analysis in a complex field setting.

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APPENDIX A: SURVEY INVITATION

Original survey invitation sent 17 Aug 2020:

Individuals invited to participate received an email invitation with the text below:

Commanders, Directors and Squadron Leaders,

As you know, one of CSAF's focus areas is "revitalizing squadrons" and over the course of his tenure we've launched numerous efforts to understand factors that inhibit as well as contribute to mission accomplishment at the squadron and installation level. With the introduction of more developmental categories for our Line of the Air Force officers it's more important than ever to understand the leadership and team processes that contribute to squadron, group and wing effectiveness and whether these developmental categories need more targeted and focused leadership development programs.

I understand how turbulent things are right now at the installation level as you balance the need to care for your Airmen and their families in the midst of a pandemic while at the same time keeping the mission going. I appreciate your willingness to consider this invitation to participate in this Air Force approved survey examining leadership competencies and team effectiveness.

You are receiving this invitation because on 31 May 2020 you were in command of a unit at the squadron level or above OR you worked directly for a squadron commander or deputy in a role such as first sergeant, deputy base civil engineer, operations officer, flight commander/director, etc.

This online survey is expected to take approximately 20 to 25 minutes to complete and the number of questions presented to you will vary based on your level of command. Your responses, which will be identifiable, will be downloaded but not viewed by our research partners at Drexel University and provided to Air Force analysts who will combine your survey responses with demographic information about you contained in the Military Personnel Data System (MilPDS) or Defense Civilian Personnel Data System (DCPDS). Air Force analysts will deidentify this combined data set by removing your name and email address and then provide this deidentified datafile to the research team to perform the analysis.

Your participation in this study will not earn you any financial or other benefits, but your time will be an investment in the future of the Air Force commanders. The results of this study will be provided, in aggregate, to AF/A1 and senior leadership for potential use in developing future commanders, directors and leaders. Participation is completely voluntary. There will be no consequences for choosing not to participate. As official Air Force sponsored research participants and for your convenience, volunteers for this study can participate and complete the survey during both duty and non-duty hours using their official government computer or their own personal device.

You have the option to stop answering questions at any point in the survey by closing the browser. If you choose to exit the survey before completion, your answers to that point will be discarded and will not be used for research. Submission of the completed survey will be interpreted as your informed consent to participate and that you affirm you are at least 18 years of age.

If you have any questions about this research, please contact the Air Force point of contact, Brig Gen (Sel) Shawn Campbell via e-mail at shawn.w.campbell.mil@mail.mil, the principal investigator, Dr. Daan van Knippenberg via e-mail at dlv39@drexel.edu, co-investigator Dr. Lauren D’Innocenzo via e-mail at led73@drexel.edu or the investigator, Theresa Carter, via e-mail at tcc72@drexel.edu. The Air Force Sponsor for this research is Brig Gen (Sel) Shawn Campbell, AF/A1, via phone at 703-692-6321 or e-mail at shawn.w.campbell.mil@mail.mil. This study has been approved by the Air Force Survey Office (SCN # AF20-090A1) as shown at the following link (<https://proxy.qualtrics.com/proxy/?url=https%3A%2F%2Fwww.my.af.mil%2Fgcscs-af%2FUSAF%2Fcontent%2Fvalid&token=75OO%2Fzje81IBtKmAB%2FkEfx07cMz7Pe8np2WJ%2FTfR0eI%3D>).

Thank you for your leadership during these challenging times and your willingness to consider participation in this study.

//Signed/BTK/18 Jun 2020//
BRIAN T. KELLY
Lieutenant General, USAF
Director, Directorate of Manpower, Personnel and Services

Follow this link to the survey:

[Take the Survey](#)

Or copy and paste the URL below into your internet browser:

https://drexel.qualtrics.com/jfe/preview/SV_2nrgcNd56ElK1W5?Q_CHL=preview

Follow this link to opt out of future emails:

[Click here to unsubscribe](#)

Clarifying e-mail sent on 20 Aug 2020 to respond to concerns invitation was a phishing attempt:

Commanders, Directors and Squadron Leaders,

On 14 Aug 2020 the e-mail below was sent to you, inviting you to participate in an Air Force approved survey examining leadership in multiteam systems. We’ve fielded a number of questions about the request’s authenticity given it was distributed by our research partners at Drexel University and not from our Air Force Survey Office (AFSO). Many were concerned this was a phishing attempt.

I wanted to reassure you that AFSO approved this survey and assigned Survey Control Number AF20-090A1 with Lt Gen Kelly as the sponsor. You can view AFSO's list of approved surveys by visiting the following

site: <https://proxy.qualtrics.com/proxy/?url=https%3A%2F%2Fwww.my.af.mil%2Fgc%2Fcontent%2Fvalid&token=7500%2Fzje81IBtKmAB%2FkEfx07cMz7Pe8np2WJ%2FTfR0el%3D>

We are working with Drexel University through an educational partnership agreement to examine leadership competencies in teams. This survey is part of that effort and we expect the results to help inform senior leaders about our leadership development programs, both now and in the future.

Finally, I'd like to clarify our intent with the second question in the survey since we've received multiple questions about this item. You'll see this question if you answer "no" to question 1. Question 2 asks "On 31 May 2020, were you assigned to a squadron or flight?" If you were assigned to a squadron in any capacity (CC, first sgt, training manager, etc.) or a numbered flight (such as "XX Contracting Flight") you should answer "Yes" to this question.

If you've already completed the survey, thank you for your time and willingness to participate in this effort. If you have not completed the survey, we appreciate your willingness to consider participating in this effort. If you have any questions or need your personalized survey link resent to you, please contact Theresa Carter at tcc72@drexel.edu.

v/r
swc

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APPENDIX B: SURVEY MEASURES

Multiteam System Effectiveness. Respondents serving in command positions at the group level and higher were asked to assess the overall performance of their subordinate units across four areas. They were asked to use the criteria in Air Force Instruction 1-2, *Commander's Responsibilities* to rate the team's overall performance (not the performance of the unit commander) during their time in command of the unit they were leading on 31 May 2020. Each item was measured using a 5-point Likert scale ranging from 1 = *Unsatisfactory* to 5 = *Outstanding*. This rating scale is familiar to Air Force leaders as it is the same scale used during unit evaluations and inspections.

1. **Executing the Mission** (ref AFI 1-2 para 3.1). Considerations in this area include *primary mission execution, AEF readiness and mission assurance command and control*.
2. **Leading People** (ref AFI 1-2 para 3.2). Considerations in this area include *communication, discipline, training, development and quality of life engagement*.
3. **Managing Resources** (ref AFI 1-2 para 3.3). Considerations in this area include *equipment, facilities and environment, guidance, and Airman's Time*.
4. **Improving the unit** (ref AFI 1-2 para 3.4). Considerations in this area include *strategic alignment, process operations, commander's inspection program and data-driven decisions*.

Intergroup Relational Identity (Rast et al., 2019). Respondents were asked to select the extent they agreed or disagreed with the following statements. Each respondent was prompted with text for [CT] or [MTS] that was tailored to their unit. [Superordinate group] is the conversational term for the MTS; i.e., 78th Air Base Wing as a superordinate group is “wing”. A 7-point Likert scale ranging from 1 = *Strongly disagree* to 7 = *Strongly agree* was used.

1. The relationship my [CT] has with the [MTS] is part of who we are.
2. The collaborative relationship between the [CT] and [MTS] is part of what makes us who we are.
3. The [CT] as a team is in part defined by our relationship with the [MTS].
4. The relationship between the [CT] and the [MTS] is part of who the [CT] and the [MTS] are.
5. The relationship between the [CT] and the [MTS] is important to what kind of [superordinate group] we are.

Leader Rhetoric (Rast et al., 2019). Respondents were asked to select the extent to which they agree or disagree with the following statements about **their immediate commander as of 31 May 2020** using a 7-point Likert scale ranging from 1 = *Strongly disagree* to 7 = *Strongly agree*. For example, if they were a flight commander, the questions referred to their squadron commander. If they were a squadron commander, the questions refer to their group commander. If they skip an echelon (i.e., they are a group commander reporting to a NAF or Center commander), the questions refer to that NAF or Center commander.

1. My commander stresses that members of our unit work with those outside our unit while preserving each unit's distinct and separate identities.
2. My commander says that our unit and other units are interdependent upon each other.
3. My commander suggests that the distinct relationship between our unit and other units in part defines their Air Force identity.
4. My commander makes clear that the collaborative relationship between our unit and other units is part of what makes us who we are.
5. My commander says that maintaining a collaborative relationship with other units while emphasizing each unit's strengths is crucial.
6. My commander emphasizes that as part of our unit's relationship with other units, each unit should develop a strong sense of identification with each other.
7. My commander believes the distinct identities between our unit and other units are important for the relationship between these units.
8. My commander thinks that part of our Air Force identity is defined by maintaining unit differences between our unit and other units.
9. My commander makes clear how our unit relates to and supports the Air Force mission.

Boundary Spanning (Faraj & Yan, 2009). Respondents were asked to read each statement and select the extent to which they perform the actions listed below. Response options used a 7-point Likert scale ranging from 1 = *Never* to 7 = *Every time*.

1. I solicit information and resources from elsewhere in and/or beyond my unit.
2. I try to influence important actors elsewhere in and/or beyond my unit on behalf of the unit and its mission.
3. I make use of relationships with others on behalf of my unit.
4. The work of my unit depends upon information and resources that I actively solicit, that is, information and resources beyond what comes through official channels.

Systems Thinking (Moore et al., 2010). Respondents were asked to read each statement and select the extent to which you perform the actions listed below. Response options used a 7-point Likert scale ranging from 1 = *Never* to 7 = *Every time*.

1. I seek everyone's view of the situation.
2. I look beyond a specific event to determine the cause of the problem.
3. I think understanding how the chain of events occur is critical.
4. I include people in my unit to find a solution.
5. I think recurring patterns are more important than any one specific event.
6. I think of the problem at hand as a series of connected issues.
7. I consider the cause and effect that is occurring in a situation.
8. I consider the relationships among coworkers in the unit.
9. I think that systems are constantly changing.
10. I propose solutions that affect the work environment, not specific individuals.
11. I keep in mind that proposed changes can affect the whole system.
12. I think more than one or two people are needed to have success.
13. I keep the mission and purpose of the unit in mind.
14. I think small changes can produce important results.
15. I consider how multiple changes affect each other.
16. I think about how different employees might be affected by the improvement.
17. I try strategies that do not rely on people's memory.
18. I recognize system problems are influenced by past events.
19. I consider the past history and culture of the unit.
20. I consider that the same action can have different effects over time, depending on the state of the system.

Identification (Mael & Ashforth, 1992). Respondents were asked to select the extent to which they agree or disagree with the following statements about your [unit] where [unit]

was flight, squadron, group, wing, NAF or Center, or MAJCOM. Response options ranged from 1 = *Strongly disagree* to 7 = *Strongly agree*.

1. When someone criticizes my **[unit]**, it feels like a personal insult.
2. I am very interested in what others think about my **[unit]**.
3. When I talk about my **[unit]**, I usually say "we" rather than "they."
4. My **[unit's]** successes are my successes.
5. When someone praises my **[unit]**, it feels like a personal compliment.
6. If a story in the media criticized my **[unit]**, I would feel embarrassed.

APPENDIX C: ARCHIVAL DATA ELEMENTS

Data Required from Personnel Systems	
Data Element	Source
Official ".mil" e-mail address	AFNET / DISA
Rank	MilPDS / DCPDS
Duty Air Force Specialty Code (DAFSC)	MilPDS / DCPDS
Command of assignment	MilPDS / DCPDS
Date of Rank (DOR)	MilPDS / DCPDS
Personnel Accounting System Code (PASCODE)	MilPDS / DCPDS
Unit	MilPDS / DCPDS
Effective Duty Date (EDD)	MilPDS / DCPDS
Below-the-Zone Promotion (Officers only)	MilPDS
Ethnic	MilPDS / DCPDS
Sex	MilPDS / DCPDS
Race	MilPDS / DCPDS
Hispanic	MilPDS / DCPDS
Core Career Field Identifier	MilPDS / DCPDS
Primary AFSC	MilPDS / DCPDS
# Days Deployed	MilPDS / DCPDS
Duty History	MilPDS / DCPDS
Joint Service (number of joint tours)	MilPDS / DCPDS
Installation Demographics -- Each installation hosting an Air Force Squadron, Group and/or Wing	
Installation Name	Note: This data (other than host unit name) will only be collected for AF Installations
Total Acreage	
Total Facilities	
Total Buildings	
Host Unit (i.e., 355 WG or Fort Hood)	
Data from Manpower Database (MPES) -- Unit Manning Document Information for each Wing, Group, Squadron and Flight	
Unit Name	MPES
PASCODE	
Total number of each AFSC broken out by officer, enlisted, and civilian	

APPENDIX D: CONDITIONAL INDIRECT AND TOTAL EFFECTS

Table 18. Conditional Indirect Effects of Leader Rhetoric on Intergroup Relational Identity with Low Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	-0.002	0.049	0.963
	Mean Systems Thinking	-0.003	0.065	0.963
	High Systems Thinking (+1 SD)	-0.004	0.081	0.963
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.003	0.064	0.963
	Mean Systems Thinking	-0.004	0.08	0.963
	High Systems Thinking (+1 SD)	-0.004	0.096	0.963
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.004	0.079	0.963
	Mean Systems Thinking	-0.004	0.095	0.963
	High Systems Thinking (+1 SD)	-0.005	0.111	0.963

Notes: N = 256. Values are standardized coefficients.

Table 19. Conditional Indirect Effects of Leader Rhetoric on Intergroup Relational Identity with Mean Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	-0.001	0.026	0.963
	Mean Systems Thinking	-0.002	0.042	0.963
	High Systems Thinking (+1 SD)	-0.003	0.058	0.963
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.002	0.041	0.963
	Mean Systems Thinking	-0.003	0.057	0.963
	High Systems Thinking (+1 SD)	-0.003	0.073	0.963
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.003	0.056	0.963
	Mean Systems Thinking	-0.003	0.072	0.963
	High Systems Thinking (+1 SD)	-0.004	0.088	0.963

Notes: N = 256. Values are standardized coefficients.

Table 20. Conditional Indirect Effects of Leader Rhetoric on Intergroup Relational Identity with High Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	0	0.003	0.965
	Mean Systems Thinking	-0.001	0.019	0.963
	High Systems Thinking (+1 SD)	-0.002	0.035	0.963
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.001	0.018	0.963
	Mean Systems Thinking	-0.002	0.034	0.963
	High Systems Thinking (+1 SD)	-0.002	0.05	0.963
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.002	0.032	0.963
	Mean Systems Thinking	-0.002	0.049	0.963
	High Systems Thinking (+1 SD)	-0.003	0.065	0.963

Notes: N = 256. Values are standardized coefficients.

Table 21. Conditional Total Effects of Leader Rhetoric on Intergroup Relational Identity with Low Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	-0.034	0.103	0.738
	Mean Systems Thinking	-0.035	0.102	0.729
	High Systems Thinking (+1 SD)	-0.036	0.103	0.728
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.035	0.102	0.73
	Mean Systems Thinking	-0.036	0.103	0.728
	High Systems Thinking (+1 SD)	-0.037	0.107	0.732
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.036	0.103	0.728
	Mean Systems Thinking	-0.036	0.106	0.732
	High Systems Thinking (+1 SD)	-0.037	0.112	0.74

Notes: N = 256. Values are standardized coefficients.

Table 22. Conditional Total Effects of Leader Rhetoric on Intergroup Relational Identity with Mean Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	-0.033	0.108	0.758
	Mean Systems Thinking	-0.034	0.104	0.743
	High Systems Thinking (+1 SD)	-0.035	0.102	0.732
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.034	0.104	0.744
	Mean Systems Thinking	-0.035	0.102	0.733
	High Systems Thinking (+1 SD)	-0.035	0.102	0.728
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.035	0.102	0.734
	Mean Systems Thinking	-0.035	0.102	0.728
	High Systems Thinking (+1 SD)	-0.036	0.105	0.729

Notes: N = 256. Values are standardized coefficients.

Table 23. Conditional Total Effects of Leader Rhetoric on Intergroup Relational Identity with High Levels of Leader Component Team Identification Combined with Leader MTS Identification and Systems Thinking

		Effect	SE	p
Low MTS Identification (-1 SD)	Low Systems Thinking (-1 SD)	-0.032	0.118	0.785
	Mean Systems Thinking	-0.033	0.111	0.766
	High Systems Thinking (+1 SD)	-0.034	0.105	0.749
Mean MTS Identification	Low Systems Thinking (-1 SD)	-0.033	0.111	0.768
	Mean Systems Thinking	-0.034	0.106	0.75
	High Systems Thinking (+1 SD)	-0.034	0.102	0.737
High MTS Identification (+1 SD)	Low Systems Thinking (-1 SD)	-0.034	0.106	0.752
	Mean Systems Thinking	-0.034	0.103	0.738
	High Systems Thinking (+1 SD)	-0.035	0.102	0.73

Notes: N = 256. Values are standardized coefficients.

Table 24. Conditional Indirect Effects of Intergroup Relational Identity on MTS Effectiveness

	Effect	SE	p
Low Boundary Spanning (-1 SD)	-0.117	0.096	0.224
Mean Boundary Spanning	-0.003	0.057	0.963
High Boundary Spanning (+1 SD)	0.111	0.07	0.11

Notes: N = 256. Values are standardized coefficients.

Table 25. Conditional Total Effects of Intergroup Relational Identity on MTS Effectiveness

	Effect	SE	p
Low Boundary Spanning (-1 SD)	-0.149	0.106	0.159
Mean Boundary Spanning	-0.035	0.102	0.733
High Boundary Spanning (+1 SD)	0.079	0.131	0.545

Notes: N = 256. Values are standardized coefficients.